



SAN DIEGO STATE UNIVERSITY

College of Engineering

Department of Mechanical Engineering

ME 499 Special Study

AMPL Review Paper

Professor: Dr. Khaled Morsi

Students: Kurtis Ramsey, Ronin Toy

Spring Semester, 2026

A. Table of Contents:

A. Table of Contents	1
B. Table of Figures	2
1. Literature Review (Kurtis)	3
On the Mechanical Performance of Closed Cell Al Alloy Foams	3
Composite Metal Foams Processed Through Powder Metallurgy	3
Heat Generation during the Fatigue of a Cellular Al Alloy	4
Effect of Aluminum Powder Agglomeration on the Foaming Al-TiH ₂ Precursors	4
Present Work Relevance	5
2. Literature Review (Ronin)	6
Metallic Foams and Their Importance	6
Powder Metallurgy Route for Closed-Cell Foams	6
Thermophysical Properties and Foam Formation	6
Electrically Assisted Heating and Current-Controlled Processing	7
Metal Foams and Electrically Driven Heating Systems	7
Relevance to the Present Work	8
3. Introduction	8
4. Project Scope	10
5. Experimental Equipment and Setup	10
Data Acquisition System	12
Experimental Procedures	13
6. Results and Discussions	16
Analysis of Effect on Heating Rate to Current (Ronin)	16
Analysis Effect of Green Density and H/W Ratio on Heating Rate; Resistivity Effect on Current (Kurtis)	29
7. Conclusion and Future Work	39
8. References	40
9. Appendix	41
DAQ Hardware and Software Setup	41

B. Table of Figures:

Figure 1: Experimental System	11
Figure 2: S.H.S Reactor	12
Figure 3: Omega DAQ	12
Figure 4: TDK Lambda Power Supply	13
Figure 5: Inside System During Heating	13
Figure 6: Powder Weighing Scale	15
Figure 7: Hydraulic Press	15
Figure 8: Test 1 Four Phase TempvsTime and Heating Rate Graphs	17
Figure 9: Test 2 Four Phase TempvsTime and Heating Rate Graphs	18
Figure 10: Test 3 Four Phase TempvsTime and Heating Rate Graphs	19
Figure 11: Test 4 Four Phase TempvsTime and Heating Rate Graphs	20
Figure 12: Test 7 Four Phase TempvsTime and Heating Rate Graphs	21
Figure 13: Test 8 Four Phase TempvsTime and Heating Rate Graphs	22
Figure 14: Test 9 Four Phase TempvsTime and Heating Rate Graphs	23
Figure 15: Test 10 Four Phase TempvsTime and Heating Rate Graphs	24
Figure 16: Test 11 Four Phase TempvsTime and Heating Rate Graphs	25
Figure 17: Post-“Foam” Samples for 250 Amps	26
Figure 18: Post-“Foam” Samples for 500 Amps	26
Figure 19: Post-Foam Samples for 750 Amps	27
Figure 20: Post-Foam Samples for 1000 Amps	27
Figure 21: Best Samples From Each Current	28
Figure 22: Chart 1 – Relative Density, Weight, H/W Ratio, Avg Top and Bottom Resistivity	30
Figure 23: Chart 2 – Relative Density vs H/W Ratio	30
Figure 24: Chart 3 – Relative Density vs Average Resistivity (Top and Bottom)	31
Figure 25: Chart 4 – Top vs Bottom Resistivity	31
Figure 26: Relative Density vs Weight	32
Figure 27: Heating Rate vs Temperature	33
Figure 28: Current vs Time	34
Figure 29: Test 7 Heating Rate and Current	35
Figure 30: Current and Heating Rate Comparison	35
Figure 31: All Foamed Samples	37

List of Tables:

Table 1: Heating Rate Analysis	16
Table 2: Archimedes Calculations	28
Table 3: Height/Width and Resistivity Measurements	29
Table 4: Archimedes Calculations	37

1. Literature Review(Kurtis):

On the Mechanical Performance of Closed Cell Al Alloy Foams:

Authors: Y. Sugimura, J. Meyer, M. Y. He, H. Bart-Smith, J. Grenstedt, and A. G. Evans (1997)

In a study on closed-cell aluminum alloy foams, Sugimura investigated three distinct types of cast alloys-ALPORAS (Type A), ALCAN (Type B), and ALULIGHT (Type C)-to determine how cell morphology and internal defects influence stiffness, yield strength, and fracture resistance. The researchers utilized micro and nanoindentation to characterize the reference properties of the cell walls, finding a wall modulus of approximately 68 GPa and a yield strength of 130 MPa for the ALPORAS variant. Through optical characterization, they identified four major categories of imperfections: curved cell walls, wall kinks or "wiggles," nodal inclusions (high-density aggregates), and wall voids. Mechanical testing in compression and flexure revealed that these imperfections, particularly the wiggles, significantly degrade performance compared to ideal models. A notable finding was the systemic asymmetry between tension and compression, where the materials exhibited higher stiffness and yield strength in tension $E_t = 1.25$ GPa than in compression $E_c = 1.05$ GPa. Fracture analysis indicated that crack growth occurs through a plastic tearing mechanism similar to thin sheets, with crack growth resistances measured at approximately 1 kJ/m². The authors concluded that while manufacturing defects currently limit these materials, they have the potential to approach the theoretical "Kelvin" solid limit if process-induced imperfections are minimized.

Composite Metal Foams Processed Through Powder Metallurgy:

Authors: B. P. Neville and A. Rabiei (2008)

Neville and Rabiei detailed the development and performance of a novel closed-cell composite metal foam (CMF) processed through a powder metallurgy (PM) technique. This method involved packing steel hollow spheres and filling the interstitial vacancies with steel powder. They tested with both carbon and stainless steel, before sintering the mixture into a solid cellular structure. The resulting foams had relative densities between 32.4% and 38.9%. Although denser than traditional aluminum foams, these CMFs showed superior compressive strength and energy absorption. Specifically, the stainless steel samples achieved a plateau strength-to-density ratio of 43.7 MPa/(g/cm³), while carbon steel ranged from 12 to 31.9 MPa/(g/cm³). Energy absorption at densification also showed promise, reaching 67.8 MJ/m³ for stainless steel. The study observed that under static loading, the material undergoes densification

through "collapse bands" that originate at the largest cells and propagate perpendicular to the loading axis. These findings demonstrate that CMFs are highly effective for applications requiring extreme energy absorption and high strength-to-weight ratios compared to traditional cellular materials.

Heat Generation during the Fatigue of a Cellular Al Alloy:

Authors: A. Rabiei, A. G. Evans, and J. W. Hutchinson (1999)

The research by Rabiei, Evans, and Hutchinson focused on the localized heat generation occurring at stress concentrators during the compression-compression fatigue of cellular aluminum alloys. By measuring the thermal flux from a notch and comparing it to a plasticity model based on the Dugdale zone, the team sought to understand the drivers behind fatigue degradation. They discovered that heat is generated through two primary processes: hysteresis within narrow cyclic plastic zones near the notch and, once the notch closes, friction along the notch faces. However, the measurements revealed that the temperature rise was too small to induce fatigue via thermal softening. Instead, the study posits that fatigue is driven by structural damage, specifically geometric softening, where cells buckle or collapse, and the gradual growth of cracks within the cell walls. This degradation is manifested as a rapid increase in strain after a stable "nucleation" period, often occurring in discrete bands one cell-size thick. This research proved that the failure of these foams under cyclic loading is a mechanical instability rather than a thermal one.

Effect of Aluminum Powder Agglomeration on the Foaming Al-TiH₂ Bulk Foamable Precursors:

Authors: Dominic Malanga, Oscar Osuna, and K. Morsi

In a critical examination of the powder metallurgy (PM) production route, Dominic, Oscar, and Dr. Morsi investigated the effect of aluminum powder agglomeration on the quality of bulk foamable precursors (BFPs). The standard PM process involves mixing a blowing agent like titanium hydride TiH₂ with aluminum powder, compacting it into a dense BFP, and heating it above the aluminum melting point to initiate foaming. The study highlighted that even partial agglomeration in the initial metal powder (a factor often overlooked in research) causes an inhomogeneous distribution of the TiH₂ particles. This lack of uniformity leads to the formation of "interconnected porosity" rather than the desired isolated, closed-cell structure. The results of their experiments showed that these defects significantly impair the final foam's structure and performance. It was concluded that addressing powder agglomeration is a vital prerequisite for future processing to ensure the production of consistent, high-quality metallic foams.

Present Work Relevance:

These four papers served as a good starting point for researching for projects involving powder metallurgy and advanced heating techniques. The study by Malanga is the most critical for the initial "pre-processing" phase; it highlights a major pitfall in precursor preparation aluminum powder agglomeration. By understanding how even partial agglomeration leads to an uneven distribution of the blowing agent and results in poor, interconnected porosity, you can establish rigorous mixing and compaction standards before ever starting an experiment. This ensures that the variables we will be testing (such as electrical current or heating rates) are not obscured by fundamental flaws in the material's initial chemical makeup.

Furthermore, these papers provide the necessary information for evaluating the "success" of a foam once it has been processed. The work of Sugimura, Neville, and Rabiei provides a framework for both morphological and mechanical characterization. Sugimura's research allows a researcher to look beyond simple density and identify specific structural defects, like cell wall "wiggles," that indicate a sub-optimal foaming process. Meanwhile, Neville and Rabiei's data on energy absorption and strength-to-density ratios provide the target performance values that any new foaming method, such as spark plasma operation, should aim to meet or exceed. Finally, Rabiei provides the long-term perspective needed for a review of powder foams, explaining how these materials eventually fail under fatigue. Together, these studies showed the raw powder inputs and the high-performance structural applications.

2. Literature Review(Ronin):

Metallic Foams and Their Importance

Metallic foams are lightweight porous materials that combine low density with high stiffness, energy absorption capability, and thermal efficiency. Closed-cell aluminum foams are especially attractive for structural and impact-resistant applications because they provide high strength-to-weight ratios while maintaining excellent crashworthiness and vibration damping characteristics. These materials have gained significant attention in aerospace, automotive, and energy applications due to their multifunctional properties.

The structure and performance of metallic foams are highly dependent on pore morphology, pore distribution, and relative density. Uniform cell structures are critical for achieving consistent mechanical and thermal properties. However, producing foams with repeatable pore structures remains a major challenge because foam expansion is strongly influenced by heating conditions and melt behavior during processing.

Powder Metallurgy Route for Closed-Cell Foams

One of the most common methods used to produce closed-cell aluminum foams is the powder metallurgy (PM) route. In this process, aluminum powder is mixed with a blowing agent, typically titanium hydride (TiH_2), compacted into a dense precursor, and heated above the melting point of aluminum. As the temperature increases, the blowing agent decomposes and releases hydrogen gas, causing the material to expand and form a porous cellular structure.

Behymer and Morsi (2023) emphasized that the success of the Powder Compact Melt Foaming Process (PCMFP) depends heavily on precursor density, heating rate, blowing agent behavior, and foaming temperature. Their review showed that improper heating conditions can lead to pore coalescence, foam collapse, and irregular cell structures. The study also highlighted that heating rate plays a critical role in synchronizing gas release from the blowing agent with softening of the aluminum matrix.

Traditional furnace heating methods often produce temperature gradients throughout the precursor because heat is transferred from the outside inward. These gradients can cause non-uniform expansion and inconsistent foam structures, motivating research into alternative heating techniques that provide faster and more controlled heating.

Thermophysical Properties and Foam Formation

The behavior of molten aluminum significantly affects foam stability and expansion behavior. Surface tension and viscosity control bubble growth, pore stability, and resistance to collapse during foaming. Park et al. (2005) investigated the thermophysical properties of

aluminum and magnesium alloys used in metal foam fabrication and found that both surface tension and viscosity decrease as temperature increases.

Their work also showed that alloying elements such as silicon, copper, and zinc influence foam behavior by altering melt properties. Increased viscosity can improve foam stability by resisting pore collapse, while reduced surface tension can enhance bubble expansion. These findings demonstrate that precise thermal control is essential during foaming because small temperature variations can significantly change melt behavior and final foam morphology.

Additionally, the study emphasized that many irregular pore structures observed in metallic foams result from insufficient control of temperature and melt dynamics during processing. This highlights the importance of investigating heating methods capable of producing rapid and uniform temperature distributions.

Electrically Assisted Heating and Current-Controlled Processing

Electrically assisted processing methods have emerged as promising alternatives to conventional furnace heating because they allow heat to be generated internally through Joule heating. Unlike furnace heating, where thermal energy diffuses slowly through the material, electrically driven heating can provide significantly faster heating rates and more localized thermal control.

Charalambous et al. (2018) investigated flash sintering using controlled current ramps and demonstrated that electrical current strongly influences heating behavior and resulting material properties. In conventional flash sintering, uncontrolled current surges create rapid power spikes that can generate localized hot spots and uneven heating. These hot spots may cause partial melting or thermal runaway within the material.

The study showed that gradually increasing current through a controlled ramping strategy reduced sudden power spikes and improved process stability. More importantly, changing the current ramp rate allowed researchers to directly influence material characteristics such as porosity, grain size, and conductivity. Faster current ramps produced higher heating rates, while slower ramps enabled more controlled thermal evolution.

These findings are highly relevant to aluminum foam processing because they demonstrate that electrical current can be used as a controllable processing parameter for modifying heating rates and material structure.

Metal Foams and Electrically Driven Heating Systems

Recent research has also explored electrically driven processing systems using conductive metal foams and graphite-based heating methods. Hong and Yang (2026) reviewed

semisacrificial-templated conversion methods in which metal foams serve both as conductive scaffolds and reactant sources. Their work demonstrated that electrically assisted systems provide rapid heating, improved temperature control, and enhanced processing efficiency compared to conventional external heating methods.

The review highlighted that conductive metallic foams are particularly well suited for electrically activated systems because they allow direct current flow through the structure, enabling internal heat generation. This reduces large temperature gradients and improves thermal uniformity throughout the material. The authors also emphasized that electrically driven methods can significantly reduce processing times while improving structural consistency.

These studies suggest that controlling electrical current during foaming may provide a way to directly control heating rate, foam expansion behavior, and final pore morphology.

Relevance to the Present Work

The literature demonstrates that heating rate is one of the most critical factors influencing closed-cell aluminum foam formation. Conventional furnace heating methods often suffer from slow heating rates and non-uniform temperature distributions, which can lead to inconsistent pore structures and foam instability.

Research on electrically assisted processing and flash sintering has shown that electrical current can significantly affect heating behavior, thermal stability, and final material properties. Controlled current ramping strategies allow heating rates to be adjusted while minimizing thermal runaway and localized overheating.

The present work builds upon these findings by investigating how changes in electrical current affect heating rates and foam formation during aluminum foam processing. By analyzing the relationship between applied current, temperature, and foam porous structure, this study aims to improve understanding of electrically driven foaming systems and contribute to the development of more controlled and repeatable aluminum foam production methods.

3. Introduction:

Closed-cell aluminum foams have attracted significant interest across aerospace, automotive, and defense industries owing to their exceptional combination of low density, high specific stiffness, and energy absorption capacity. These materials are most commonly produced via a powder metallurgy route, in which aluminum powder is blended with a foaming agent, typically titanium hydride (TiH_2), compacted into a dense precursor, and heated until decomposition of the foaming agent drives cellular expansion within the metal matrix. Despite the relative maturity of this approach, conventional furnace-based heating imposes considerable

limitations on the process, including slow and poorly controlled heating rates, non-uniform temperature distributions across the precursor, and inconsistent foam microstructures, all of which compromise the repeatability and quality of the final product.

Self-Propagating High-Temperature Synthesis (SHS) offers a compelling alternative thermal processing strategy. By harnessing highly exothermic reactions that sustain themselves once ignited, SHS is capable of generating rapid heating rates and extreme local temperatures without continuous external energy input, characteristics that are inherently difficult to achieve through furnace-based methods. However, conventional SHS implementations do suffer from limited control over temperature distribution and reaction stability, which has restricted their application to precision processing tasks.

The present work is motivated by the potential to address the shortcomings of both approaches through their integration. Through the use of SHS within the powder metallurgy foaming route, this study aims to improve the controllability and repeatability of closed-cell aluminum foam production, ultimately enabling more consistent and tailored foam microstructures.

To address these limitations, this project explores an electrically driven heating approach based on spark plasma conduction. Rather than relying on external furnace heating, high electrical current is passed directly through conductive components surrounding the sample, generating heat internally via resistive dissipation. This method enables significantly faster and more controllable heating rates compared to conventional approaches, offering improved precision over temperature evolution throughout the foaming process.

The SHS reactor employed in this work was originally designed and built by a previous graduate student at the Advanced Materials Processing Laboratory (AMPL) for the synthesis of nickel-aluminide compounds. For the purposes of this project, the reactor was substantially modified to support spark plasma foaming of aluminum. These modifications included mechanical and electrical redesign of the reactor configuration, as well as the integration of a data acquisition system capable of real-time monitoring of temperature, current, and voltage, enabling systematic characterisation of the heating process.

The experimental program pursued three principal objectives: first, to characterise the influence of current output on sample heating rate under fixed voltage conditions; second, to investigate the effects of green density and height-to-width ratio on heating behaviour; and third, to examine the relationship between sample resistivity and the resulting current response. Together, these objectives aim to establish a fundamental understanding of the process parameters governing spark plasma foaming of aluminum within the modified SHS reactor.

4. Project Scope:

At the outset of this project, six core objectives were established to guide the experimental program. These comprised: first, commissioning the data acquisition system (DAQ) and associated software to enable simultaneous recording of two temperature channels, current, and voltage; second, conducting multiple experimental runs with thermocouples positioned within both the graphite tooling and the sample; third, determining sample heating rates through time-differential analysis of the recorded temperature data; fourth, correlating heating rate with current output; fifth, investigating the effects of green density and height-to-width ratio on heating behaviour; and sixth, examining the relationship between sample resistivity and the resulting current response.

To support the first objective, a detailed experimental procedure was developed for the setup and operation of the Omega DAQ system, utilising InstaCal and TracerDAQ software. This procedure documented the configuration required to simultaneously acquire thermocouple temperature measurements alongside current and voltage outputs across multiple channels. A second procedure was subsequently developed to govern the setup and execution of the broader experimental programme, providing standardised guidelines for producing foam samples under varying conditions and ensuring consistent data collection across all runs.

The findings of this project are intended to help with the selection of optimal green compact parameters and power supply settings for the production of high-quality closed-cell aluminum foams in future projects within Dr. Morsi's laboratory. While the results are directly applicable to Al-TiH₂ foams, the relationships established between processing parameters and heating behaviour may offer broader utility for other mixtures involving other materials as well.

5. Experimental Equipment and Set Up:

The figure below shows our full experimental setup along with the connections from the S.H.S reactor, TDK Lambda power supply, DAQ, TracerDAQ pro, thermocouples, and current and voltage wirings. In order to make sure the crucible was heated evenly, graphite sheets were placed between the top and bottom of the crucible and copper rods to help get more uniform electrical current distribution. The S.H.S reactor was powered by the TDK lambda power supply. The TDK-Lambda power supply was used to pump high current into the reactor, by connecting it to both a four-phase wall power source and the reactor's copper rods. Getting the wiring right was a major focus, in order to ensure every connection was tight, properly grounded, and safe to handle the high power. This setup was the backbone of the project, if the electricity wasn't perfectly stable, the heating wouldn't work correctly, and the whole experiment could fail. The DAQ and the tracerDAQ pro were used to help capture current and heating rate data while the system was running. This data was recorded by thermocouple readings that were installed into the omega DAQ system which is how real time readings were recorded while the crucible was

being heated. For spark plasma heating the system passes electrical current through copper rods, through the graphite sheets and crucible, which heats and foams the aluminum, TIH_2 powder green compact placed inside the crucible. In this experiment graphite is a resistive heating element that applies a high temperature heating environment.

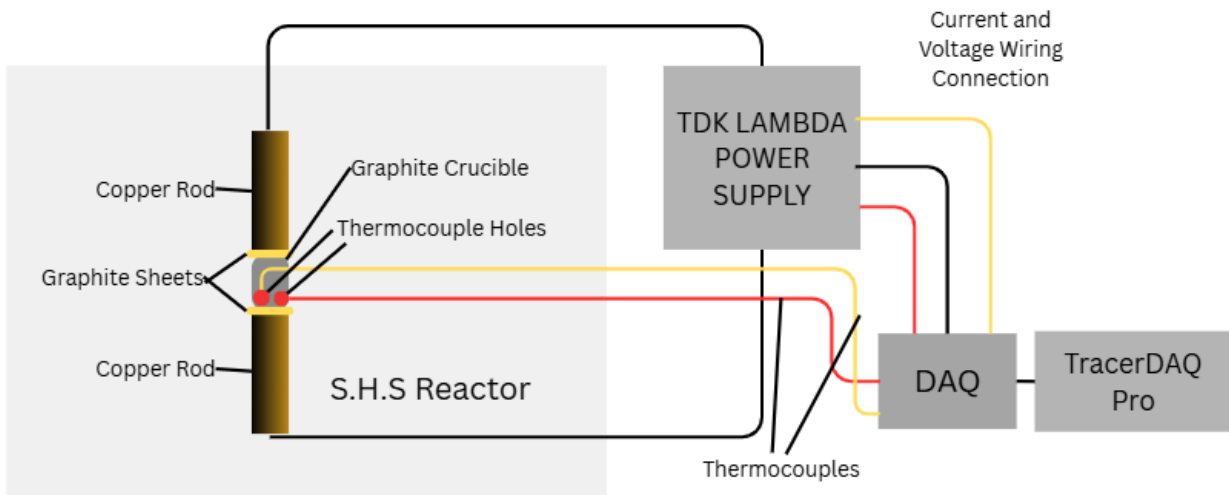


Figure 1:Experimental System



Figure 2: S.H.S Reactor

Data Acquisition System:

In order to obtain data during our experiments, a comprehensive Data Acquisition (DAQ) system was implemented. This setup featured two thermocouples placed at critical points within the reactor (specifically the graphite crucible and the sample area) to monitor heat levels in real-time. By connecting these sensors to an Omega DAQ unit, temperature changes as they occurred were able to be precisely tracked.

Beyond thermal data, we also integrated the system with the power supply's output pins to capture voltage and current measurements. Using InstaCal for hardware configuration and TracerDAQ for data logging, we could view simultaneous strip charts that displayed the relationship between electrical input and thermal output. This allowed us to record data at high sampling rates.

During initial testing, it was discovered that the high-current environment created significant electrical noise, which threatened the accuracy of our readings. This led to a focus on refining our grounding and wiring strategies to ensure cleaner data. Eventually it was found out that the system produced significantly less noise when plugged into the lab computer with older software, rather than one of our laptops. Ultimately, this DAQ integration is the key to performing a quantitative analysis of the foaming process, providing the hard data needed for our experiments.



Figure 3: Omega DAQ



Figure 4: TDK Lambda Power Supply



Figure 5: Inside System During Heating

Experimental Procedures:

This project's procedure was built on careful testing to make sure everything worked exactly as planned.

First, a pressed aluminum TiH₂ mixed green compact sample is placed inside a graphite container, with thin layers of graphite paper on the top and bottom of the graphite crucible to help the electricity flow better. The crucible is then tucked securely between two copper rods. Before turning anything on, the thermocouples are connected to the holes in the crucible, the power is turned on, then we double-check every connection to ensure safety.

When the power starts, the current is slowly turned up, causing the graphite to heat up so much that it glows. This intense heat melts the aluminum inside and triggers the foaming agent, which makes the metal expand into a foam. Once the foam has formed, the power is turned off, and the system cools down before the final foam is taken out.

The test we ran during this experiment consisted of 4 different cylindrical green compacts. Two of the green compact samples that were tested during our experiments consisted of 3.5 gram aluminum, 1.5% TiH₂ mixed powders at 92%, and 98% relative density. The other two samples consisted of 4.5 gram aluminum, 1.5% TiH₂ mixed powders at 92%, and 98% relative density. For determining the effect of effect on heating rate to current, only samples of 3.5 gram aluminum, 1.5% TiH₂ mixed powders at 98% relative density were used. For Analysis effect of green Density and height to width ratio on heating rate and analysis of resistivity effect on current, all four green compact samples were tested.

All 4 of the samples that were tested were made using the same process. First we would weigh out close to the desired weight of the compact we intended to make using the powder weight scale. We then used the hydraulic press to make the cylindrical green compacts using the double punch method. Unlike the single punch hydraulic press, in a double-punch setup, pressure is applied from both the top and the bottom simultaneously. This "squeezes" the powder from both sides at once. This significantly reduces the effects of friction and creates a much more uniform density throughout the entire part. The cylindrical punch diameter was roughly 19mm in diameter. The height of the sample varied depending on which type of sample was being made. After formation of the green compact, it was extracted and sanded down to get rid of any impurities, and ensure an even, uniform surface on the green compact. Finally, the sample and weighed and ready for testing and other measurement analysis like resistivity, and relative density of the sample.



Figure 6: Powder Weighing Scale



Figure 7: Hydraulic Press

Results and Discussions, Analysis of Effect on Heating Rate to Current(Ronin):

The following results explore how the manipulation of the current output of the power supply affects the heating rate of the sample. Four different current outputs (250 amps, 500 amps, 750 amps, and 1000amps) with a standard voltage of 5 volts were used on 9 different samples, all about 3.4-3.5g with a relative density around 98%. With these results it is able to be determined which current output should be used for the best foams to be produced. It is important to note that Test 5 and 6 are omitted from the results due to crucibles cracking during the experiment resulting in inaccurate results.

Table 1: Heating Rate Analysis

Test	Start Temp (°C)	Peak Temp (°C)	Time to Peak (s)	Avg Rate (°C/s)	Avg Rate (°C/min)
Test1 250 Amps	36.6	251.6	325.2	0.661	39.7
Test2 500 Amps	42.0	485.5	60.6	7.318	439.1
Test3 750 Amps	64.6	766.8	47.8	14.690	881.4
Test4 1000 Amps	55.1	819.1	30.2	25.264	1515.8
Test7 750 Amps	21.1	763.9	40.3	18.441	1106.5
Test8 1000 Amps	43.2	799.7	26.2	28.920	1735.2
Test9 500 Amps	43.4	630.7	129.8	4.524	271.4
Test10 500 Amps	32.0	483.0	319.2	1.413	84.8
Test11 250 Amps	56.0	260.8	103.7	1.974	118.5



Figure 8: Test 1 Four Phase TempvsTime and Heating Rate Graphs

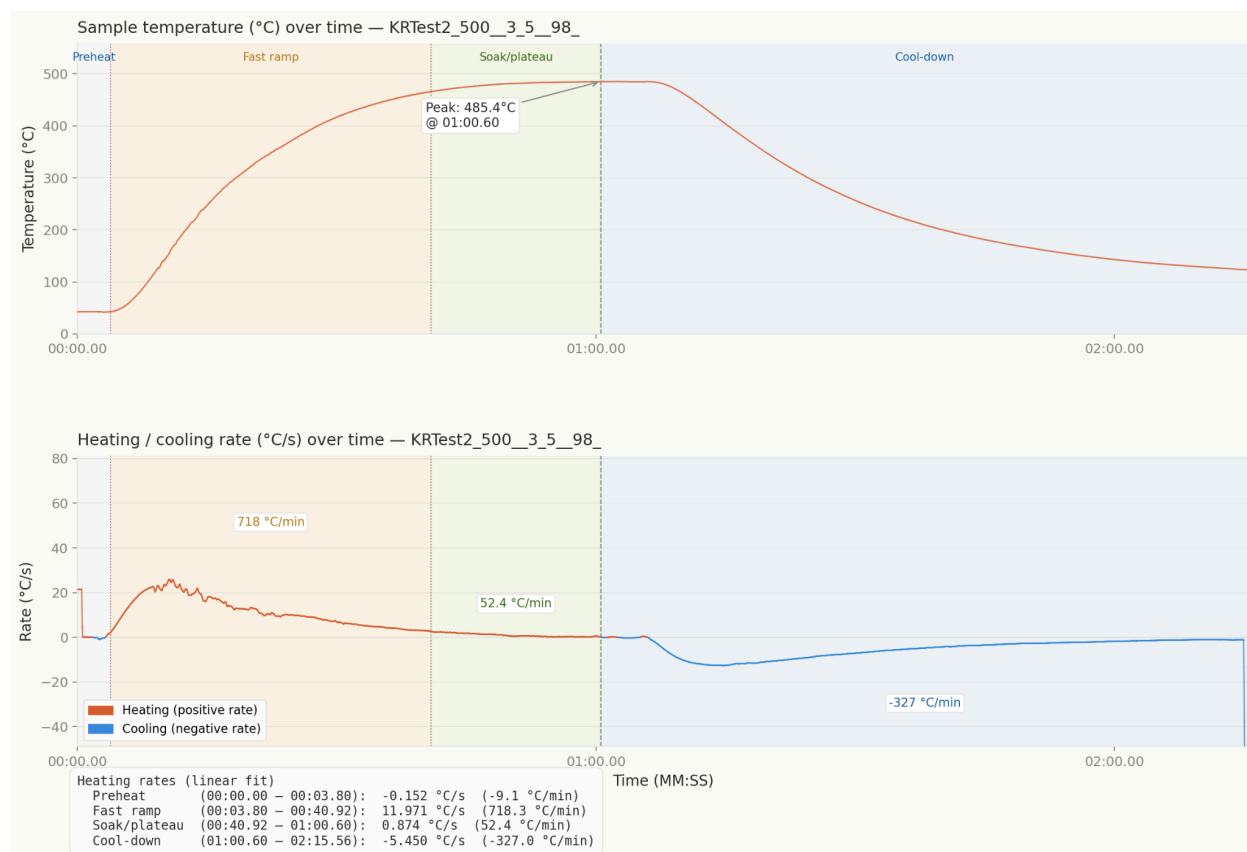


Figure 9: Test 2 Four Phase TempvsTime and Heating Rate Graphs

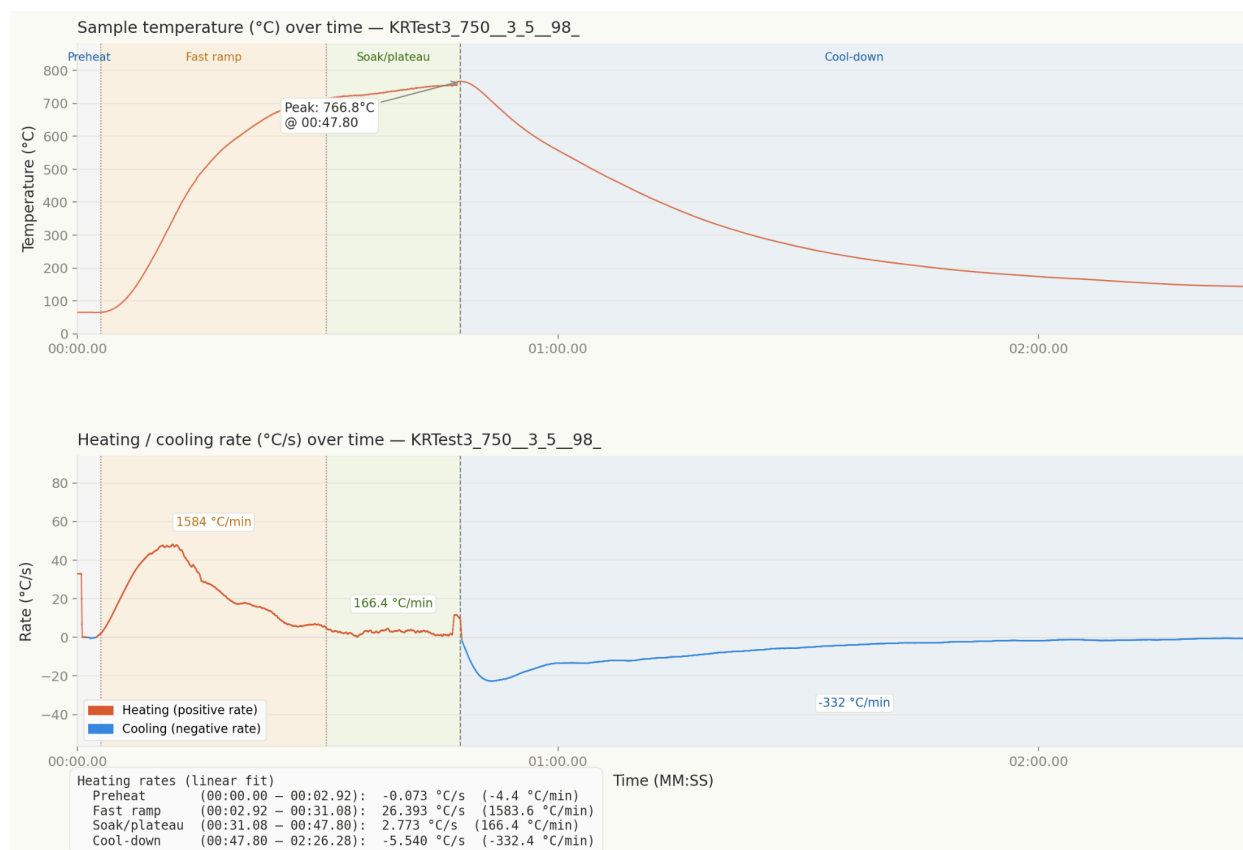


Figure 10: Test 3 Four Phase TempvsTime and Heating Rate Graphs

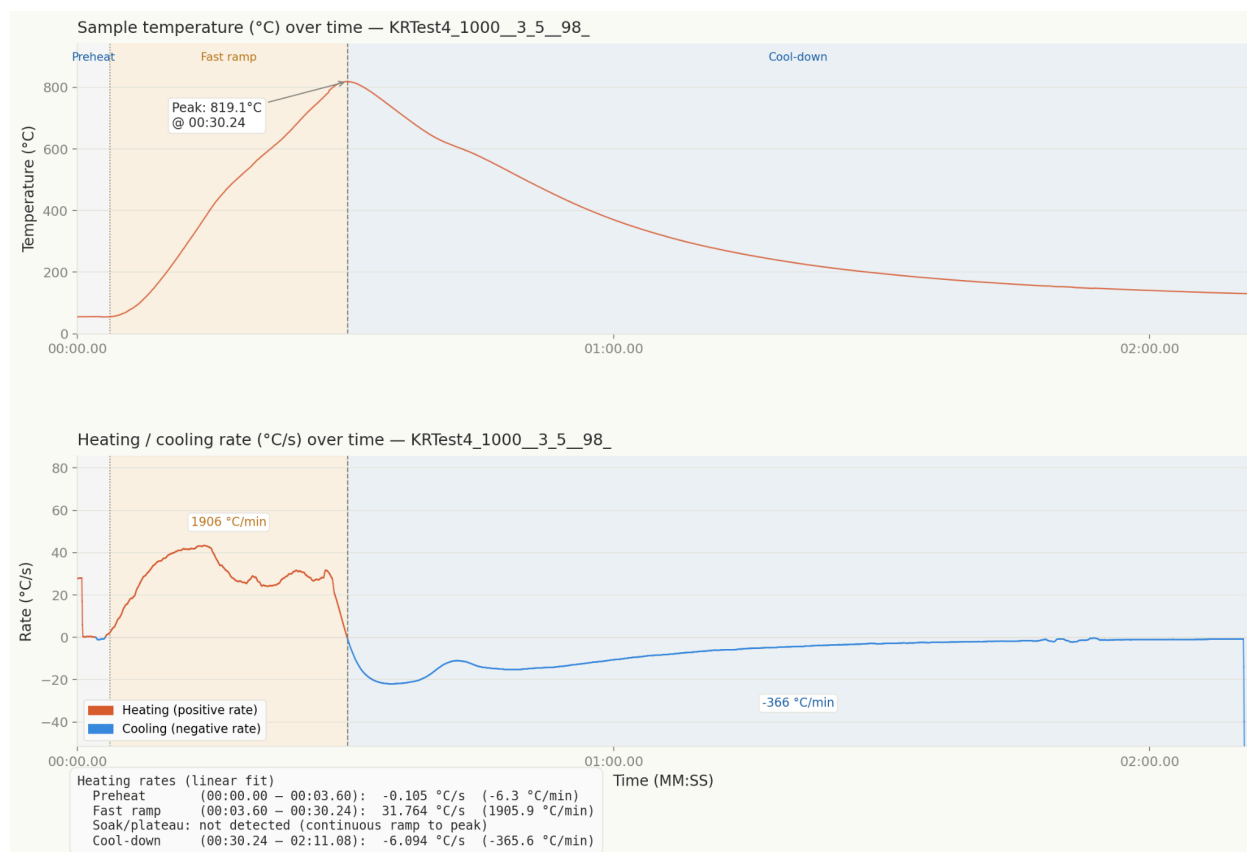


Figure 11: Test 4 Four Phase TempvsTime and Heating Rate Graphs



Figure 12: Test 7 Four Phase TempvsTime and Heating Rate Graphs



Figure 13: Test 8 Four Phase TempvsTime and Heating Rate Graphs

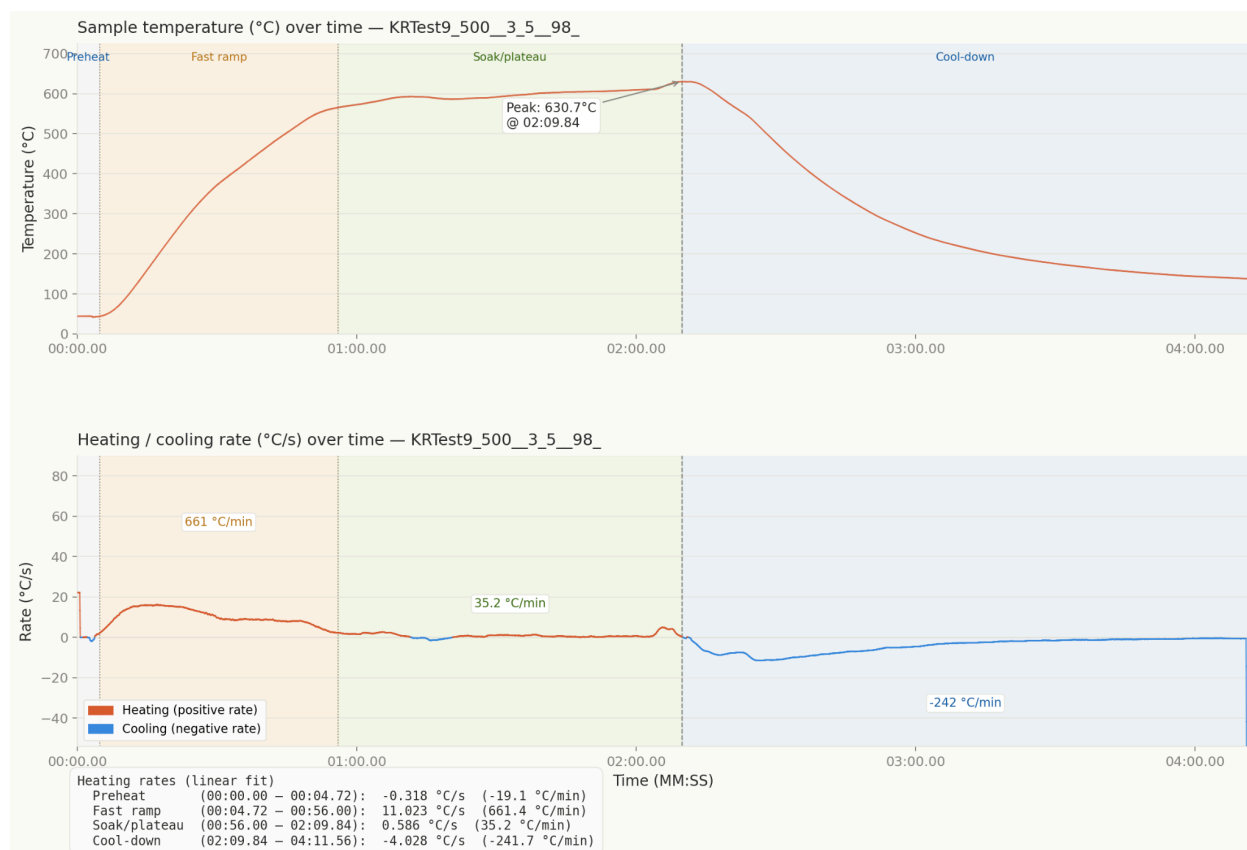


Figure 14: Test 9 Four Phase TempvsTime and Heating Rate Graphs

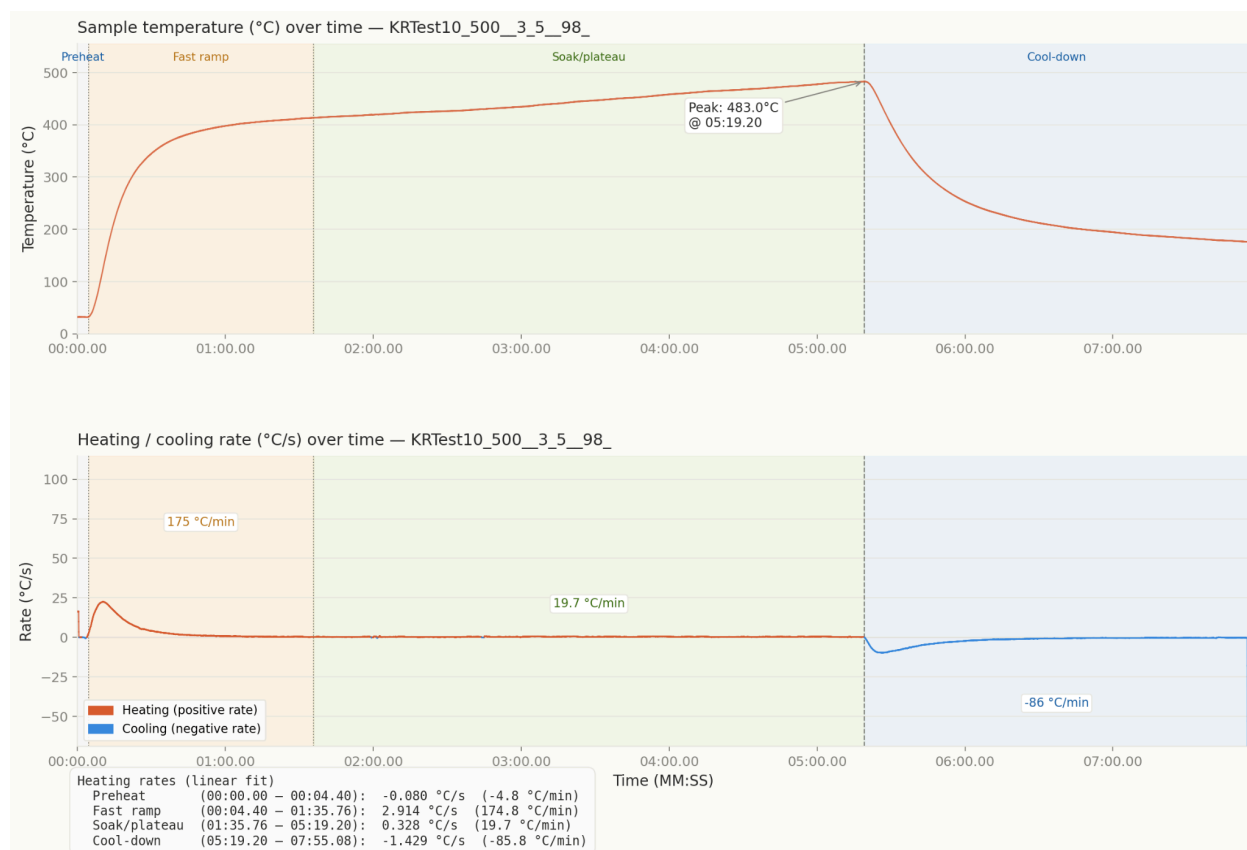


Figure 15: Test 10 Four Phase TempvsTime and Heating Rate Graphs

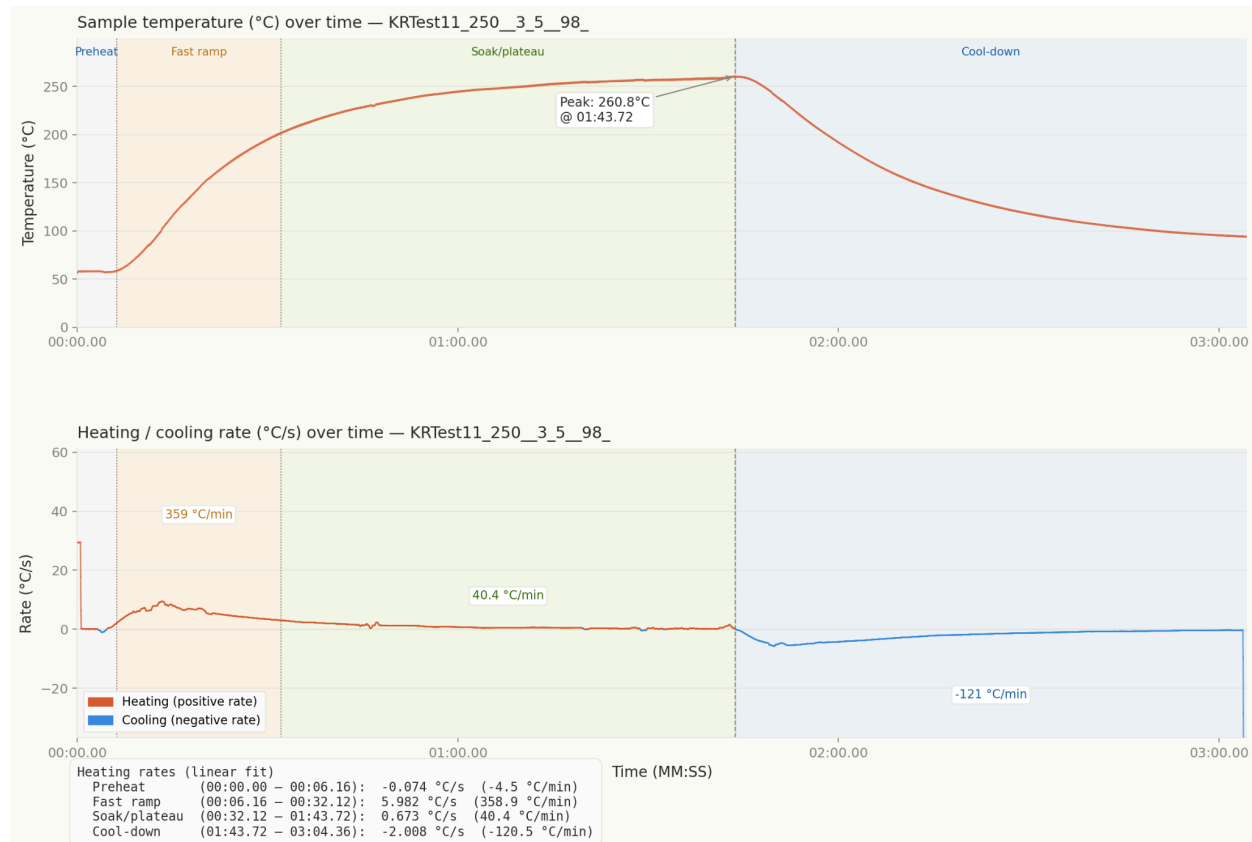


Figure 16: Test 11 Four Phase TempvsTime and Heating Rate Graphs

After looking at purely just the heating rates and temperatures it can be determined that when running the power supply at 5 volts, only the 750 Amp and 1000 Amp current outputs were able to reach high enough temperatures for the sample to foam. The only other current output to be able to foam a sample was the 500 Amp output. However, only one of the three 500 Amp tests was able to achieve foaming. This is also the reason this current value is the only one to be performed three times. Since, the foaming result wasn't able to be replicated again, it can be assumed that the 500 Amp output is too inconsistent for creating foams. Between the 750 Amp and 1000 Amp outputs it was determined that the 750 Amp output had the best heating rate for creating foams. This is because the two 750 Amp tests had an average heating rate of 993.95 °C/min, which allowed the sample to reach a peak temperature of 750-850 °C within a 90 second time window. Also, the sample temperature plateaued around 750-850 °C which helped ensure the sample didn't burn or the foam to collapse. On the other hand, the 1000 Amp test had an average heating rate of 1625.5 °C/min, which did allow the sample to reach the desired temperature within the desired time frame. However, since it heated so much faster and was able to exceed the desired sample temperature, there was more of a risk of it burning the sample and the foam collapsing. Also, there is a slight delay between the thermocouple and DAQ readings, so between human error, the delay, and manually turning off the system, the risk of the sample exceeding the desired temperature increases with a faster heating rate.

Images of samples:

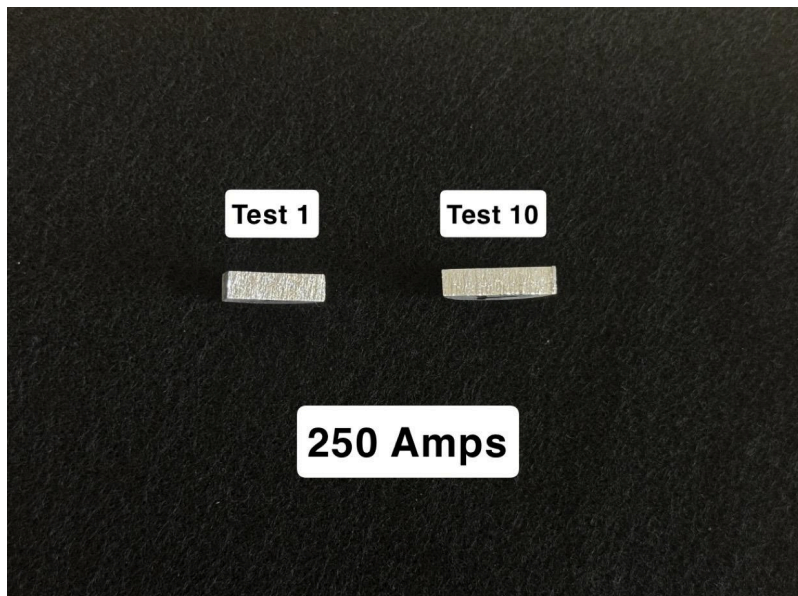


Figure 17: Post- “Foam” Samples for 250 Amps

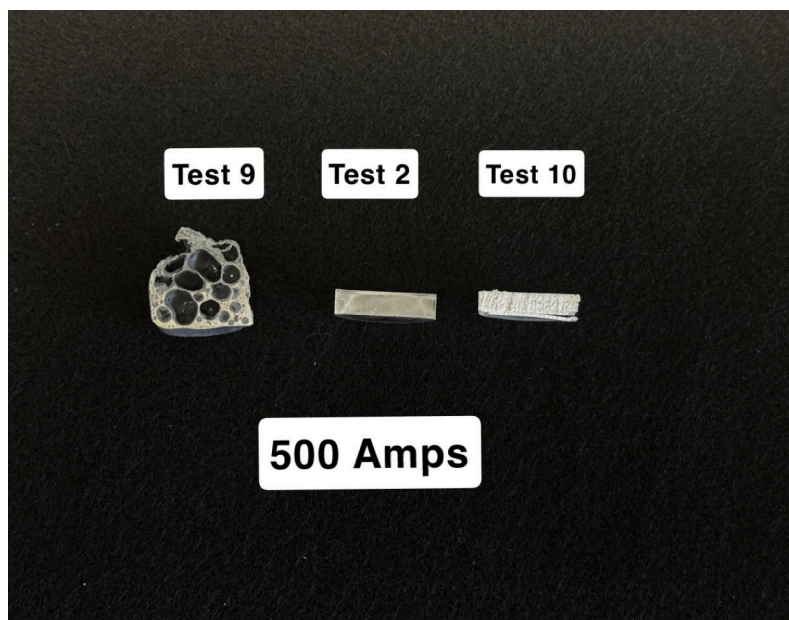


Figure 18: Post- “Foam” Samples for 500 Amps

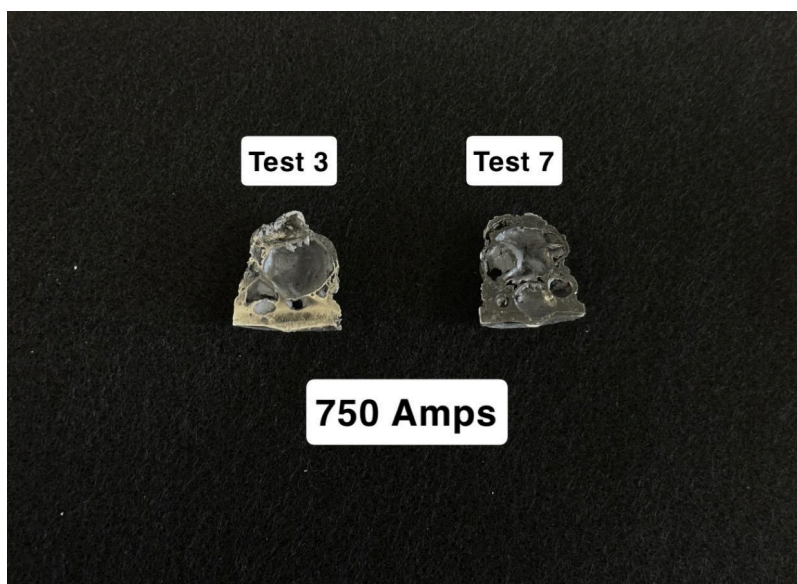


Figure 19: Post-Foam Samples for 750 Amps

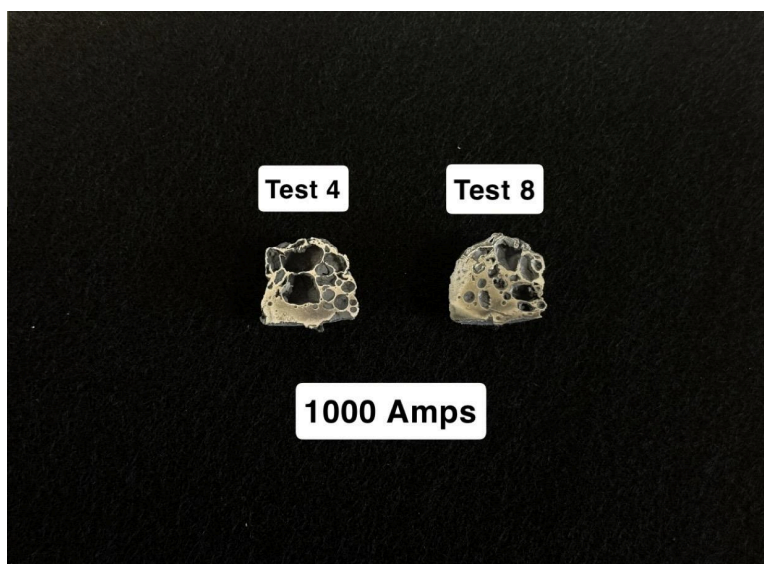


Figure 20: Post-Foam Samples for 1000 Amps

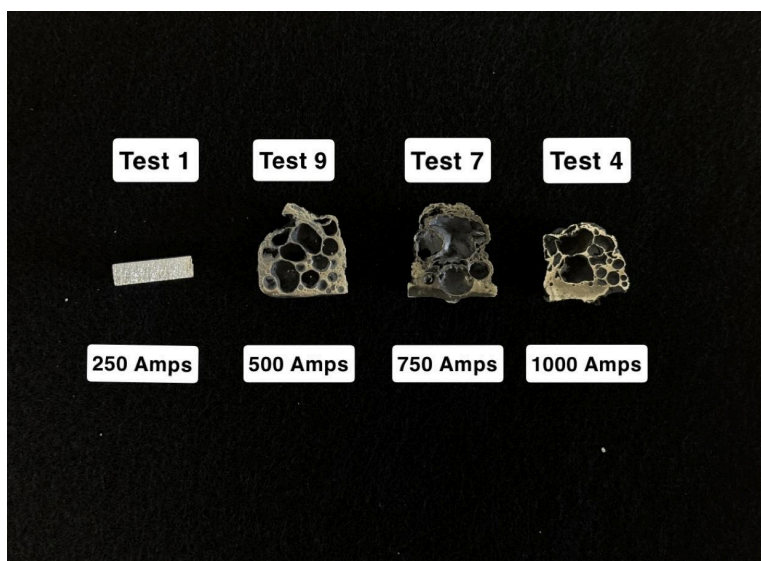


Figure 21: Best Samples From Each Current

Table 2: Archimedes Calculations

Test #	Relative Density	Dry Weight	Suspended Weight	Soaked Weight	Bulk Density (g/cm ³)	Total Porosity	Open (apparent) Porosity	Closed Porosity
1	99.19%	3.3395	2.015	3.3493	2.5028	7.30%	0.73%	6.57%
2	96.88%	3.392	1.9254	3.4234	2.2644	16.14%	2.10%	14.04%
3	99.10%	3.4306	0.9624	4.6652	0.9265	65.69%	33.34%	32.34%
4	99.12%	3.425	0.7531	4.2377	0.9829	63.60%	23.32%	40.27%
7	98.88%	3.4205	0.769	4.6365	0.8844	67.24%	31.44%	35.80%
8	99.63%	3.4252	0.6289	4.0663	0.9965	63.09%	18.65%	44.44%
9	97.50%	3.368	0.2228	4.2152	0.8436	68.76%	21.22%	47.54%
10	97.34%	3.3505	2.0642	3.3787	2.5489	5.60%	2.15%	3.45%
11	96.91%	3.3657	2.0893	3.4007	2.5665	4.94%	2.67%	2.28%

Looking at the porosity percentage for all the samples, other than the one anomaly foam from 500 Amps (Test 9), the 750 Amp foams (Test 3 and 7) out perform the 1000 Amp foams (Test 4 and 8). However, when looking at the closed porosity percentage, which means the pores didn't collapse into each other and interconnect, the 750 Amp samples performed worse. This is further shown in the pictures of the foams cross-sectioned. The 750 Amp samples look like the pores "merged" into bigger pores/bubbles. On the other hand, the 1000 Amps sample and one 500 Amp sample had a distribution of more uniform sized pores.

Overall, although the 750 Amp output seems the best for foaming based on heating rate alone, it appears that it might not create the best uniformed foams after further inspection post foam. One reason for the pores not being as uniform as the 1000 Amp and 500 Amp sample might be due to it being held in the plateau region for too long at high temperatures. This might allow the original pores to expand for too long and eventually merge with each other. An argument could be made that the 500 Amp sample was in the plateau region for longer, but the temperatures in its plateau region weren't as high and not in the foaming temperature range. In the end, the one 500 Amp test seemed to produce the best foam, but wasn't able to be replicated consistently and with its heating rate it took too long to reach the desired sample temperature. Although the 750 Amp samples seemed to have the worst pore uniformity and distribution, the issue can possibly be fixed by turning off the power supply sooner and letting the sample be in the plateau region for a shorter amount of time. It is believed if the issue of the pore sizes and uniformity is fixed, the 750 Amp current output would be the desired option because of its heating rates.

Results and Discussions, Analysis Effect of Green Density and Height to Width Ratio on Heating Rate, Analysis of Resistivity Effect on Current(Kurtis):

Table 3: Height/ Width, Resistivity Measurements

Test	Relative Density	Weight	Width	Height	Top Resistivity 1	Top Resistivity 2	Top Resistivity 3	Bottom Resistivity 1	Bottom Resistivity 2	Bottom Resistivity 3
3	99.10%	3.4306	19.067	4.493	0.009	0.009	0.011	0.016	0.014	0.007
7	98.88%	3.4205	19.026 7	4.513	0.011	0.043	0.081	0.009	0.016	0.011
12	92.50%	3.4196	19.05	4.82	0.054	0.028	0.019	0.009	0.014	0.021
13	91.69%	4.3071	19.035	6.13	0.007	0.007	0.007	0.026	0.021	0.009
14	97.48%	4.3407	19.02	5.853	0.009	0.009	0.007	0.011	0.011	0.007
15	92.33%	3.4611	19.056 7	4.9	0.04	0.097	0.047	0.014	0.021	0.023
16	93.24%	4.2492	19.03	5.98	0.009	0.007	0.011	0.007	0.011	0.019
17	97.71%	4.3749	19.23	5.81	0.009	0.011	0.016	0.011	0.026	0.007

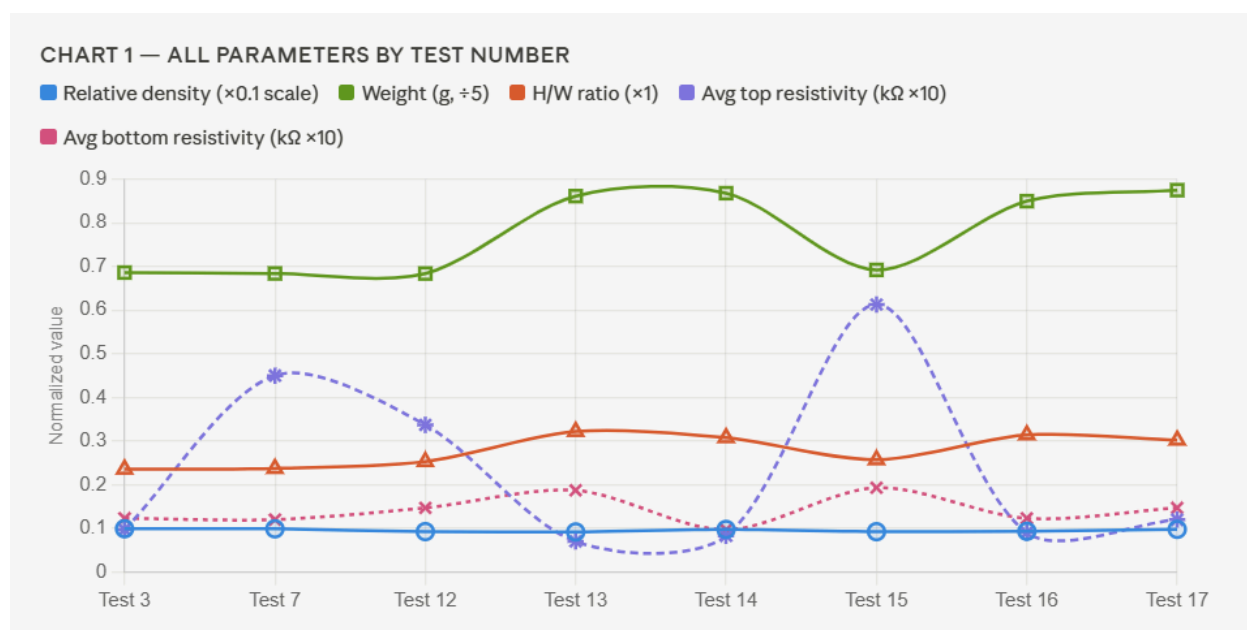


Figure 22: Chart 1- Relative Density, Weight, H/W Ratio, Avg Top and Bottom Resistivity

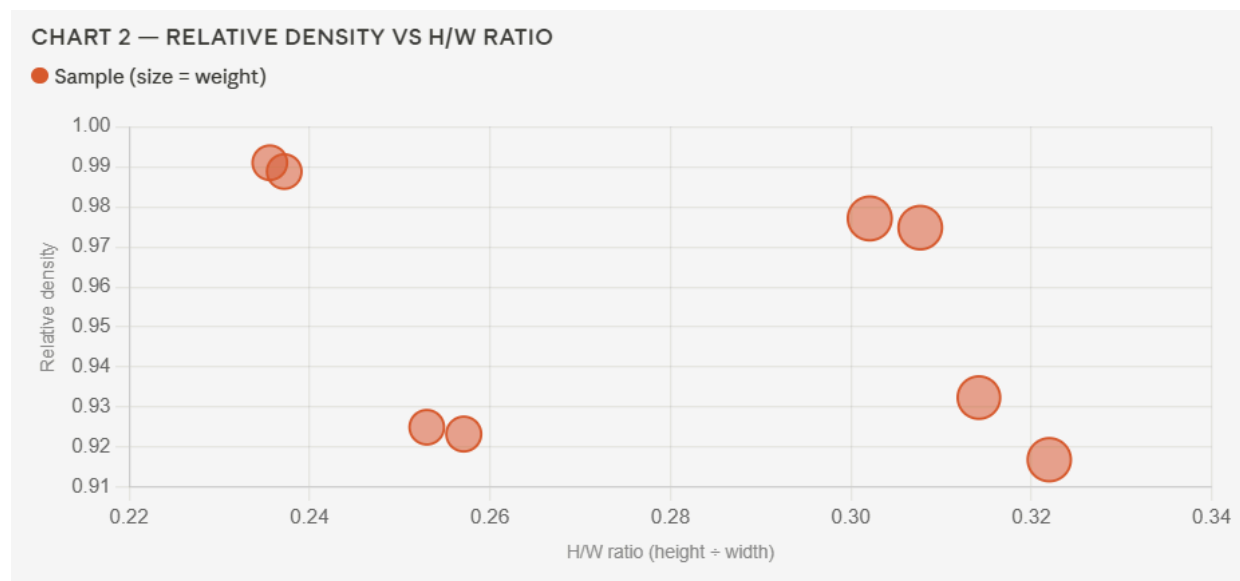


Figure 23: Chart 2- Relative Density vs H/W Ratio- The taller samples (higher H/W) tend to cluster around lower relative densities (0.917–0.932), suggesting that taller geometries may be less densely sintered. The denser samples (Tests 3, 7, 14, 17) have lower H/W ratios.

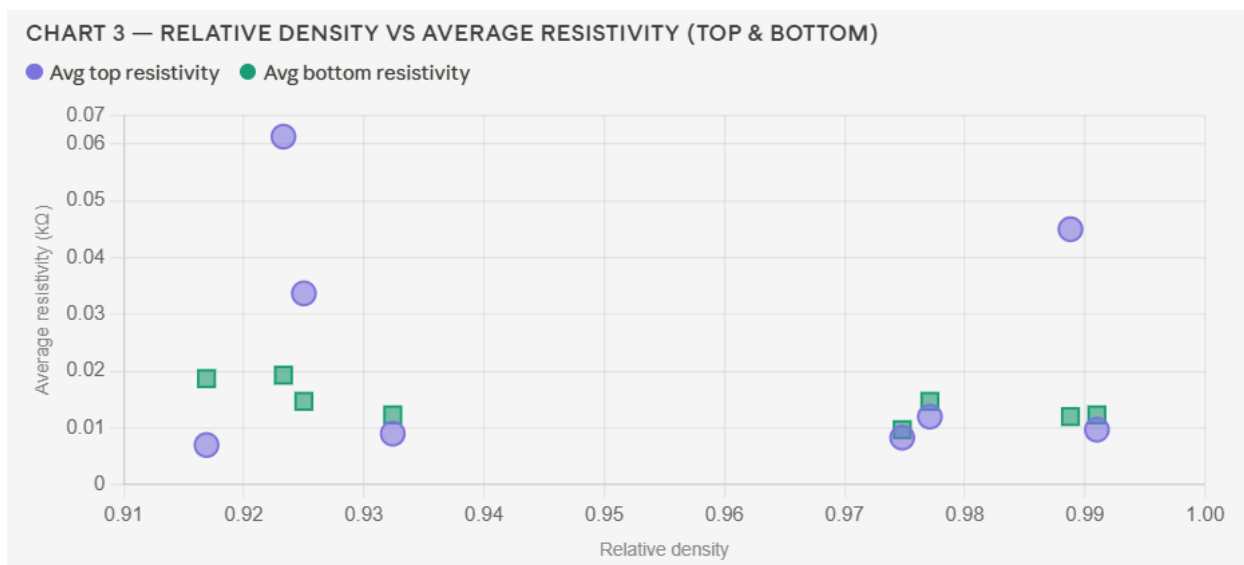


Figure 24: Chart 3-Relative Density vs Average Resistivity(Top and Bottom)- Tests 12 and 15 — both with lower relative density (~ 0.92 – 0.93) — show significantly higher top resistivity, meaning lower-density samples resist current flow more. Bottom resistivity stays relatively flat. High-density samples (Tests 3, 7, 17) cluster near the low-resistivity end.

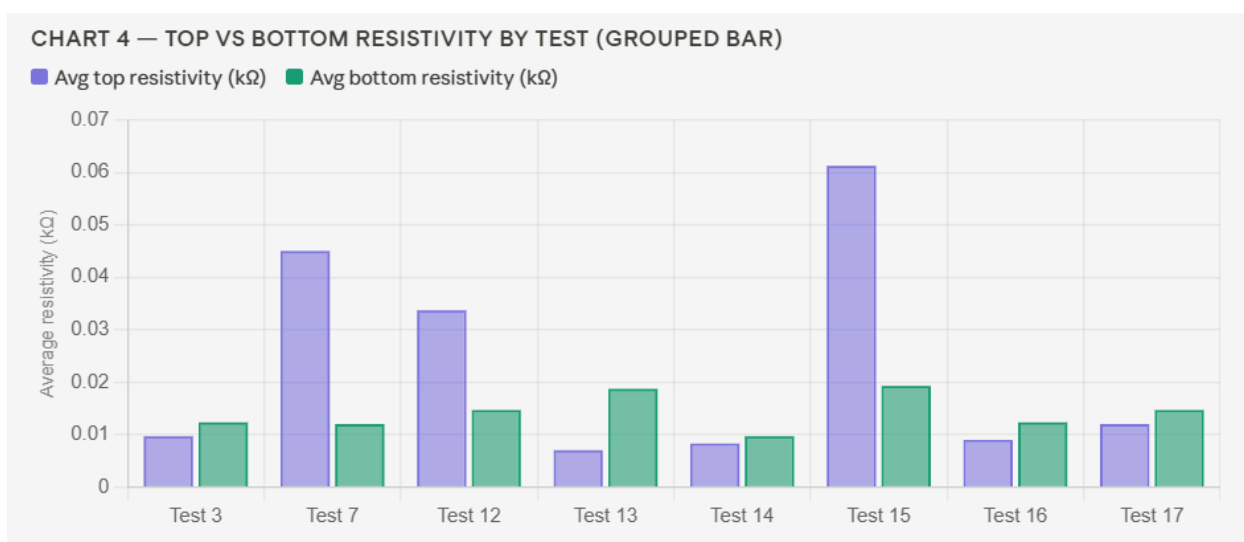


Figure 25: Chart 4-Top vs Bottom Resistivity- Tests 7 and 15 stand out with much higher top-surface resistivity than bottom, suggesting non-uniform sintering or electrode contact issues that would directly affect local heating rates.

For current and heating rate analysis the lower relative density correlates with higher resistivity (especially on top), taller samples (high H/W) tend to be less dense, and the top-to-bottom resistivity asymmetry in some tests warrants investigating whether heating is uniform through the sample thickness.

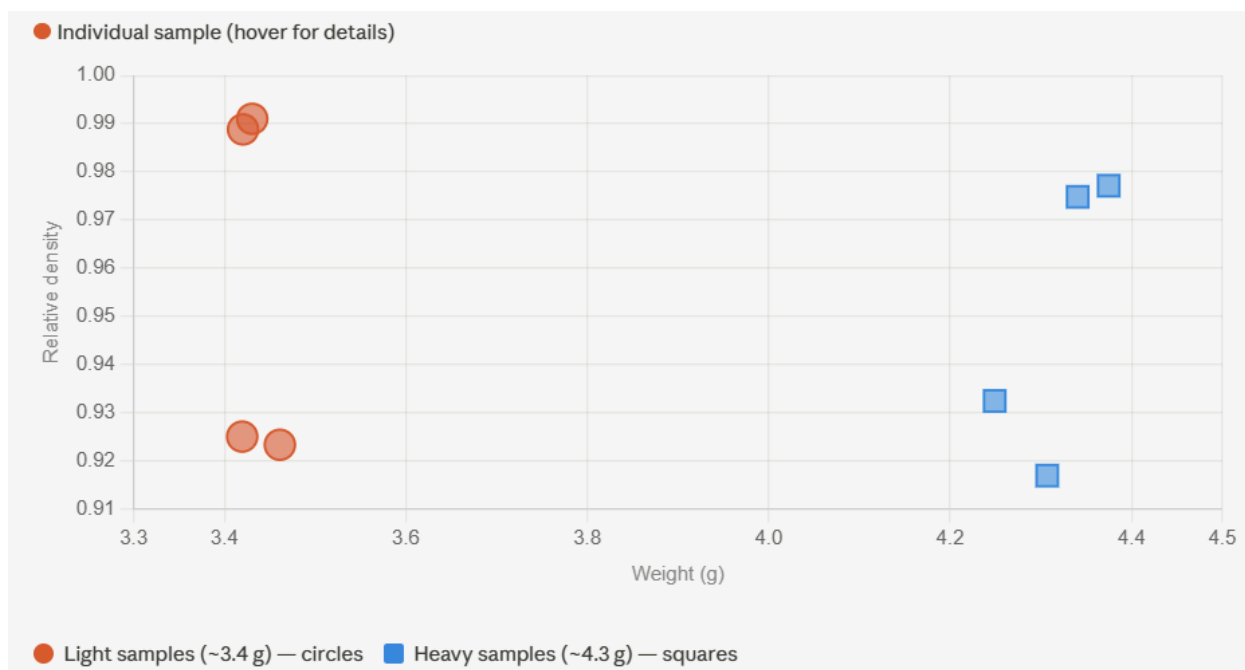


Figure 26: Relative Density Vs Weight

Light samples (~3.4g): Tests 3 and 7 are highly dense (≈ 0.989 – 0.991), while tests 12 and 15 drop to ≈ 0.923 – 0.925 . Despite nearly identical weights. This suggests powder packing or sintering conditions, not mass, are driving density differences here.

Heavy samples (~4.3g): Similarly split, tests 14 and 17 are well-densified (≈ 0.975 – 0.977), while Tests 13 and 16 are notably lower (≈ 0.917 – 0.932).

Heating Rate vs. Temperature — Individual Tests

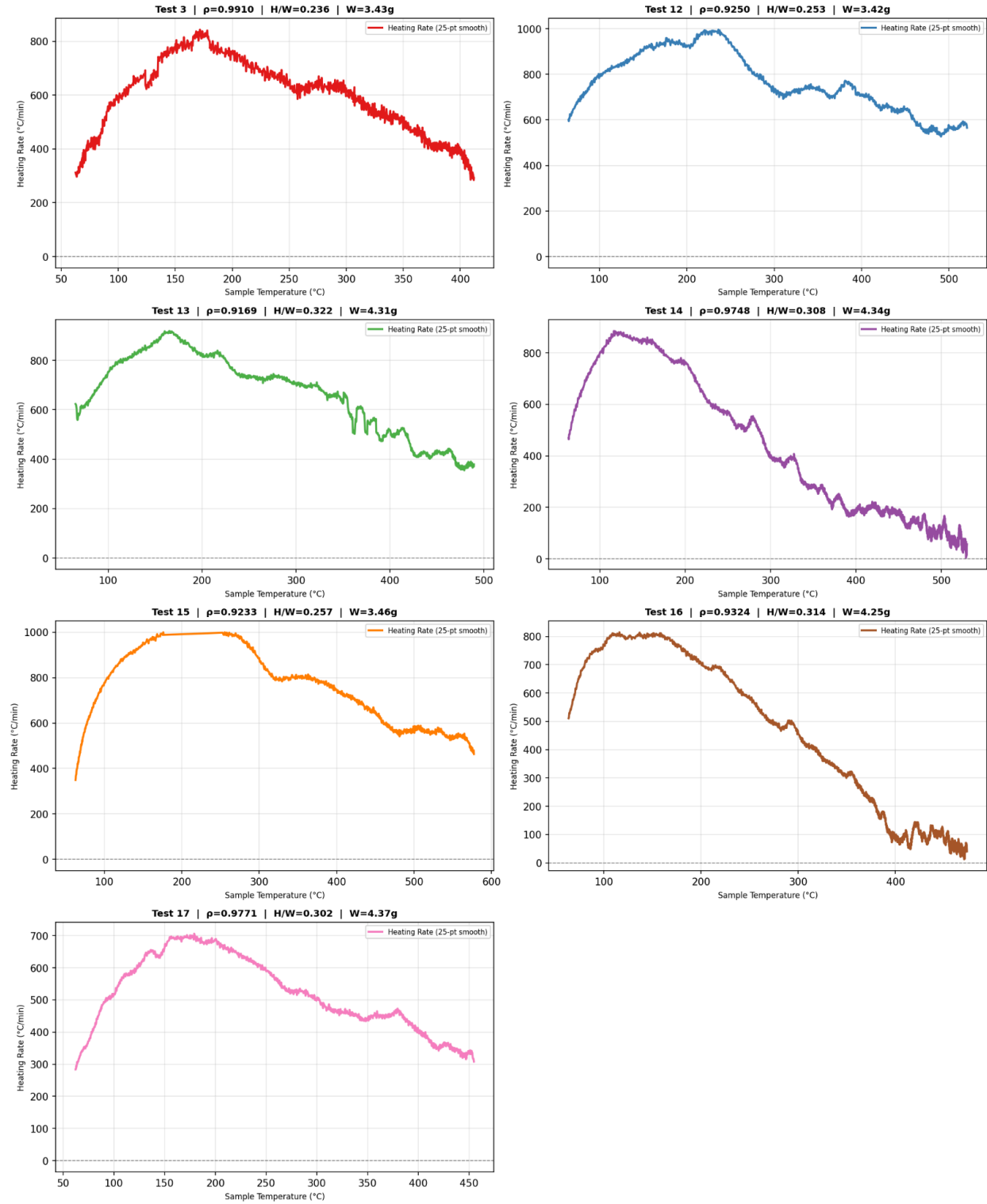


Figure 27: Heating Rate vs Temperature

Current vs. Time — Individual Tests

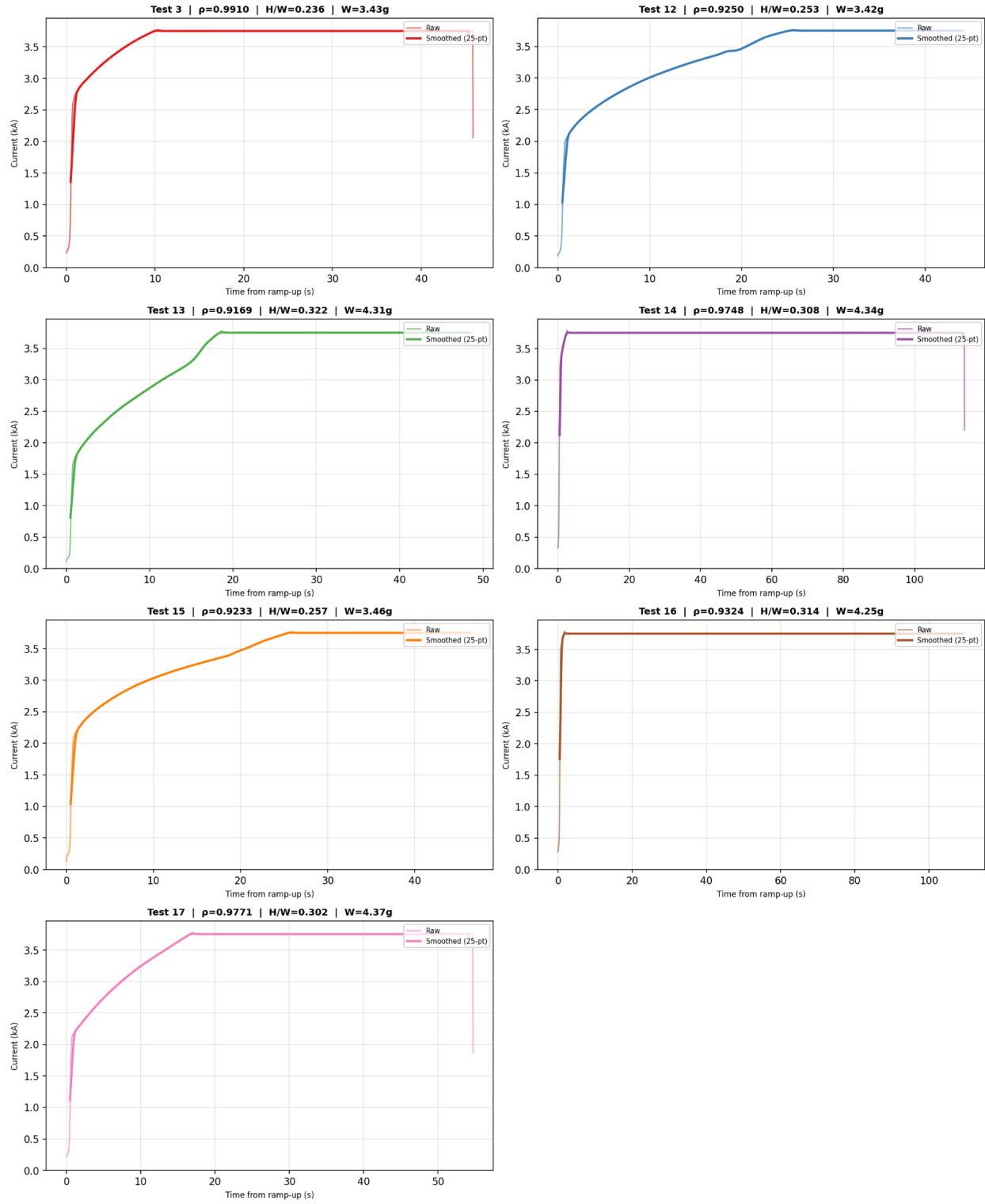


Figure 28: Current Vs Time

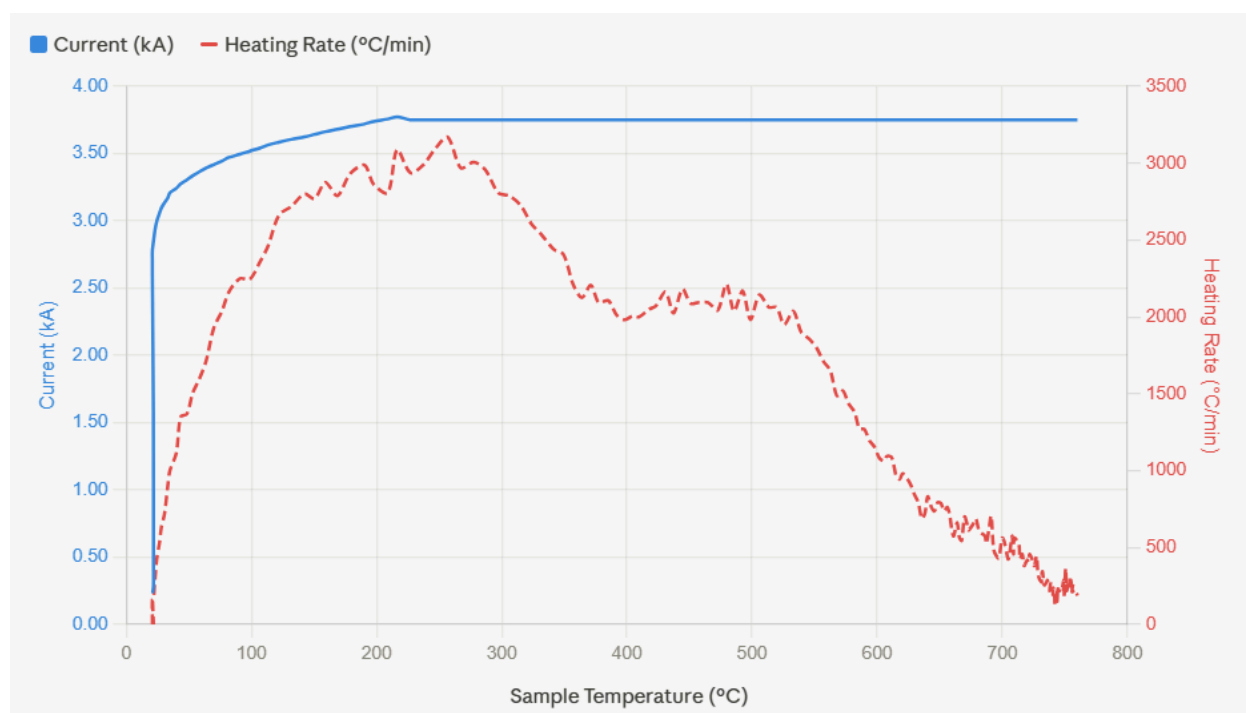


Figure 29: Test 7 Heating Rate and Current

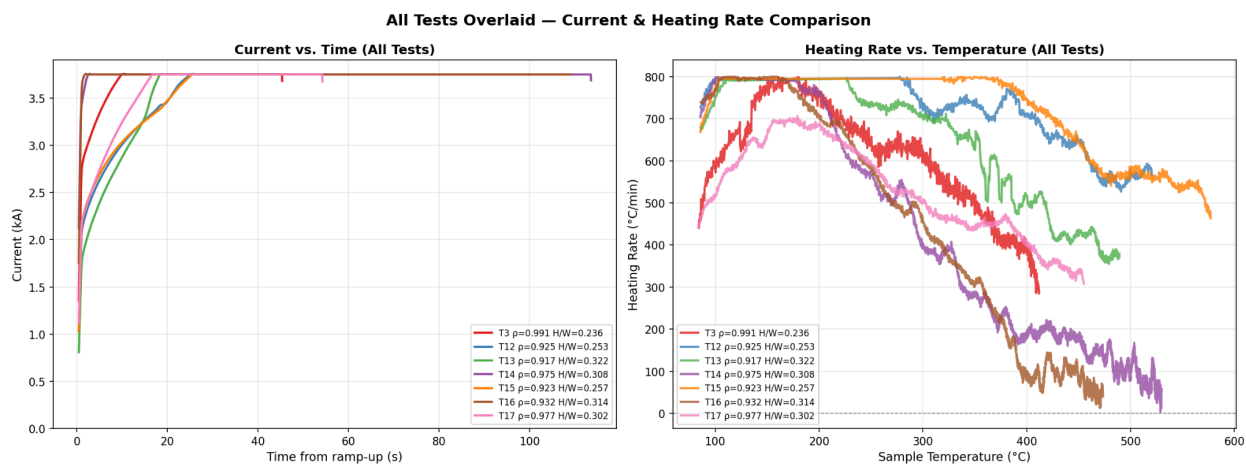


Figure 30: Current and Heating Rate Comparison

Current:

The steady-state current is nearly identical across all seven tests (3.740–3.751 kA), which is understandable because this is a controlled sintering process where the power supply targets a fixed current setpoint. The slight variations (~ 0.01 kA) are within instrument noise. Relative density, weight, H/W ratio, and resistivity had no meaningful effect on the applied current, as these are all determined by the machine's control loop rather than sample properties.

Heating Rate: Heating rate is where the most notable differences appear according to the data.

Height/Width Ratio has the strongest apparent relationship with heating rate. Tests 14 (H/W = 0.308) and 16 (H/W = 0.314) both show dramatically lower mean heating rates (~ 210 – 220 °C/min) and much longer ramp durations (109–114 s), compared to tests with lower H/W ratios (0.236–0.257), which all heated rapidly at 570 – 789 °C/min in under 50 seconds. This suggests taller samples relative to their width require significantly more time to heat through, likely due to greater thermal mass and longer heat conduction path, even when the current is identical.

Weight correlates with slower heating, but this is very similar with the H/W ratio. The heavier samples (13, 14, 16, 17 at ~ 4.3 g) include both the slow-heating tests (14, 16) and moderate ones (13, 17). Weight alone doesn't cleanly predict heating rate once H/W is accounted for - This was different for test 13 (4.307 g, H/W = 0.322) which heated relatively fast despite being tall and heavy, possibly because of its lower density.

Relative density shows a weak negative trend with heating rate. The lower-density samples (12, 13, 15) tended to heat faster than higher-density ones at similar geometry. This may suggest that lower-density (more porous) samples have less thermal mass per unit volume and possibly higher local resistive heating per unit cross-section.

Resistivity had no clear or consistent effect on either heating rate or current. The values are all very small (0.007–0.097 k Ω) and the variation doesn't correlate meaningfully with heating behavior, suggesting contact resistivity at these magnitudes is not the limiting factor.

Overall, H/W ratio appears to be the dominant geometric factor controlling heating rate. Weight matters secondarily through thermal mass. Relative density shows a mild trend (lower density = faster heating). Current is machine-controlled and essentially constant.

Table 4: Archimedes Calculations

Test #	Relative Density	Dry Weight	Suspended Weight	Soaked Weight	Bulk Density (g/cm ³)	Total Porosity	Open (apparent) Porosity	Closed Porosity
3	99.10%	3.4306	0.9624	4.6652	0.9265	65.69%	33.34%	32.34%
7	98.88%	3.4205	0.769	4.6365	0.8844	67.24%	31.44%	35.80%
12	92.50%	3.4196	1.9663	3.6382	2.0453	24.25%	13.07%	11.17%
13	91.69%	4.3071	2.5183	4.5344	2.1364	20.88%	11.27%	9.60%
14	97.48%	4.3407	2.0511	5.4595	1.2735	52.83%	32.82%	20.01%
15	92.33%	3.4611	1.9603	3.6925	1.9981	26.00%	13.36%	12.64%
16	93.24%	4.2492	2.1808	5.1512	1.4305	47.02%	30.37%	16.65%
17	97.71%	4.3749	1.4925	5.4684	1.1004	59.25%	27.50%	31.74%

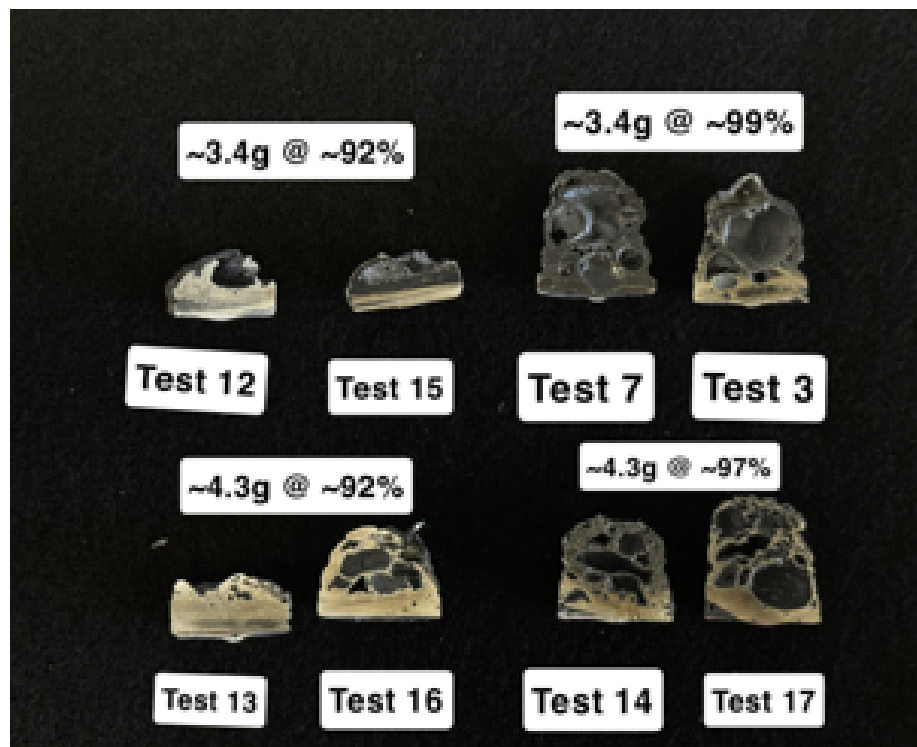


Figure 31: All Foamed Samples

After reviewing the Archimedes calculations, the best-performing samples are those from high relative density compacts that also exhibit the highest total porosity values, particularly those above 50%. This combination suggests successful foam formation where significant gas expansion occurred without compromising the overall structural integrity of the sample. By Archimedes analysis, Test 17 stands out as the best-performing sample, achieving a relative density of 97.71% with a total porosity of 59.25% and, critically, a closed-to-open pore ratio that favors closed porosity (31.74% closed vs. 27.50% open). The dominance of closed pores in Test 17 indicates that the pores remained discrete and did not coalesce or interconnect during the foaming process, which is the ideal outcome for a high-quality metallic foam. Cross-sectional analysis further confirms this, showing that Test 17 produced the most uniform and evenly distributed pore structure among all samples presented, with pores that appear spherical and well-separated throughout the bulk.

Test 3 also performed strongly in terms of relative density (99.10%) and closed porosity (32.34%), and actually exhibited the highest closed porosity of any sample in the dataset. However, its total porosity of 65.69% — while the highest recorded — paired with a high open porosity of 33.34% suggests that some pore interconnection did occur, likely as a result of over-sintering during the plateau region. Since Test 3 and Test 7 were held at temperature for a longer duration, the samples likely began to experience pore collapse and coalescence, converting what would have been isolated closed pores into an interconnected open network. Test 7 shows a similar trend, with 31.44% open porosity despite a high relative density of 98.88%, reinforcing the interpretation that extended dwell time at temperature is detrimental to pore stability.

Among the heavier samples—Tests 13, 14, 16, and 17, which all fall in the ~4.3g weight group—Test 17 and Test 14 show the most promise, though for different reasons. Test 14 achieved a relatively high density of 97.48% and the highest total porosity in the heavy group at 52.83%, but its open porosity of 32.82% substantially outweighs its closed porosity of 20.01%, suggesting significant pore collapse and interconnection. This is not reflected well by the raw density number alone, which underscores the importance of the open-to-closed pore ratio as a quality metric. Test 13, while having the lowest relative density (91.69%) and lowest total porosity (20.88%) of all samples, does show a favorable closed-to-open ratio (9.60% closed vs. 11.27% open), but the absolute porosity values are too low to consider it a successful foam. Tests 12, 15, and 16 fall into a mid-range category. Test 16 (93.24% relative density, 47.02% total porosity) and Test 15 (92.33%, 26.00% total porosity) both show moderate foaming but with open porosity dominating, indicating incomplete or unstable foam formation. Test 12 (92.50%, 24.25% total porosity) similarly underperforms in total porosity despite its relatively high open-to-closed ratio.

In summary, ranked by overall foaming quality based on the combination of high relative density, high total porosity, and favorable closed-to-open pore ratio, the samples perform as

follows: Test 17 > Test 3 > Test 7 > Test 14 > Test 16 > Test 15 > Test 12 > Test 13. Future work should focus on replicating the compaction and sintering conditions of Test 17 while experimenting with reduced plateau dwell times to determine whether the pore closure behavior seen in Tests 3 and 7 can be preserved without triggering collapse.

Conclusion and Future Work:

This study investigated the use of an electrically driven spark plasma conduction approach for closed-cell aluminum foam production within a modified SHS reactor. Through systematic experimentation across eight green compact samples of varying density, geometry, and resistivity, as well as testing across four current output levels — 250, 500, 750, and 1000 amperes at a fixed voltage of 5 volts — several key process-property relationships were established.

With respect to current output, only the 750 and 1000 ampere settings consistently achieved the temperatures necessary for foaming, while the 500 ampere output produced foaming in only one of three attempts and was deemed too inconsistent for reliable use. The 750 ampere output yielded an average heating rate of approximately 994 °C/min, enabling the sample to reach the target temperature range of 750–850 °C within a 90 second window and maintaining a stable plateau that minimised the risk of sample degradation. The 1000 ampere output, while capable of reaching the desired temperature range, produced heating rates averaging 1626 °C/min- increasing the risk of overheating and foam collapse due to the reduced margin for operator response. Post-foaming analysis revealed that although the 750 ampere samples exhibited higher total porosity, their closed porosity was comparatively lower, with evidence of pore coalescence attributed to extended dwell times within the plateau region. The 1000 ampere and single successful 500 ampere samples produced more uniform pore size distributions. It is believed that reducing plateau dwell time at 750 amperes would mitigate pore merging, making it the preferred current setting for future work.

Relative density, height-to-width ratio, and electrical resistivity were all found to further influence heating rate and current response during processing. Higher density compacts exhibited lower and more symmetric resistivities between top and bottom surfaces, producing more uniform heating profiles. Taller samples with higher H/W ratios tended toward lower relative densities, suggesting that aspect ratio affects powder consolidation quality and, consequently, current distribution during the foaming run.

Archimedes analysis confirmed that the optimal indicator of foam quality is the combination of high relative density and a favourable closed-to-open pore ratio. Test 17 produced the best result overall, with a relative density of 97.71%, total porosity of 59.25%, and

closed porosity exceeding open porosity, yielding the most uniform pore distribution of all samples tested. Tests 3 and 7, despite strong density values, showed evidence of pore coalescence attributed to extended plateau dwell times, reinforcing dwell time as a critical and sensitive processing variable.

These findings provide actionable guidance for future work, recommending compact geometries with lower H/W ratios, higher green densities, minimised plateau dwell times, and the use of pre-sintering resistivity as a predictive screening tool. While derived from an Al–TiH₂ system, the relationships established are expected to offer broader applicability to other powder metallurgy foaming systems processed under electrically driven heating conditions.

8. References(Ronin):

- [1] M. Hong and S. Yang, “Semisacrificial-templated direct conversion of metal foams: synthesis methods and applications,” *Coord. Chem. Rev.*, vol. 564, p. 218072, 2026.
- [2] S.-H. Park, Y.-S. Um, C.-H. Kum, and B.-Y. Hur, “Thermophysical properties of Al and Mg alloys for metal foam fabrication,” *Colloids Surf. A Physicochem. Eng. Asp.*, vol. 263, pp. 280–283, 2005.
- [3] N. Behymer and K. Morsi, “Review: Closed-cell metallic foams produced via powder metallurgy,” *Metals*, vol. 13, no. 5, p. 959, 2023.
- [4] H. Charalambous, S. K. Jha, K. H. Christian, R. T. Lay, and T. Tsakalakos, “Flash sintering using controlled current ramp,” *J. Eur. Ceram. Soc.*, vol. 38, pp. 3689–3693, 2018.
- [5] Omega Engineering, “Introduction to thermocouples and temperature measurement,” Omega Technical Reference, 2020.
- [6] Measurement Computing, “TracerDAQ Pro Data Acquisition Software User’s Guide,” 2021.

References(Kurtis):

- [1] Y. Sugimura, J. Meyer, M. Y. He, H. Bart-Smith, J. Grenstedt, and A. G. Evans, “On the mechanical performance of closed cell Al alloy foams,” *Acta Materialia*, vol. 45, no. 12, pp. 5245–5259, Dec. 1997, doi: 10.1016/S1359-6454(97)00148-1.
- [2] B. P. Neville and A. Rabiei, “Composite metal foams processed through powder metallurgy,” *Materials & Design*, vol. 29, no. 2, pp. 388–396, Feb. 2008, doi: 10.1016/j.matdes.2007.01.026.
- [3] A. Rabiei, A. G. Evans, and J. W. Hutchinson, “Heat Generation during the Fatigue of a Cellular Al Alloy,” *Metallurgical and Materials Transactions A*, vol. 31, no. 4, pp. 1129–1136, Apr. 2000.

[4] D. Malanga, O. Osuna, and K. Morsi, "Effect of Aluminum Powder Agglomeration on the Foaming Al-TiH₂ Bulk Foamable Precursors," Journal of Manufacturing and Materials Processing, 2024.

9. Appendix:

DAQ Hardware and Software Setup

Download these two Softwares:

InstaCal:

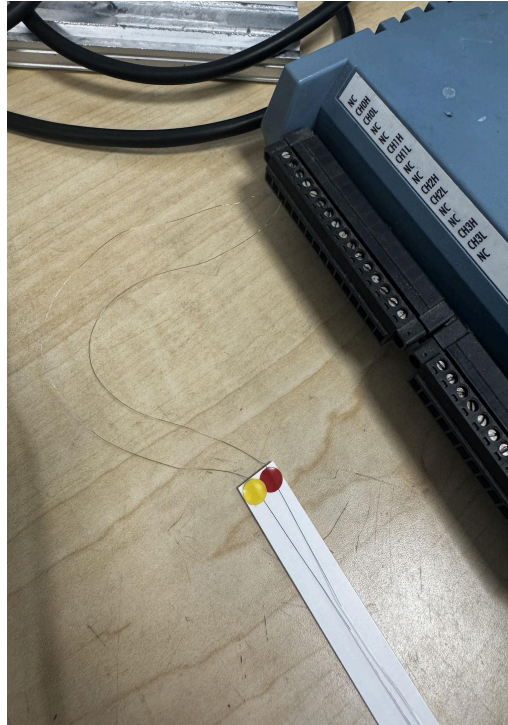
https://digilent.com/reference/software/instacal/start?srsId=AfmBOooDNtgEfBDEADVWc9CAS0V0_hygVmgnon48bo1qyskM5mbvezWB

TracerDAQ or TracerDAQ Pro (Both Free):

https://digilent.com/reference/software/tracerdaq/start?srsId=AfmBOoq-paMaN8IMl1xEiawhine-rS-gFo0TrQyJO24zYW_lq6VrG7P6

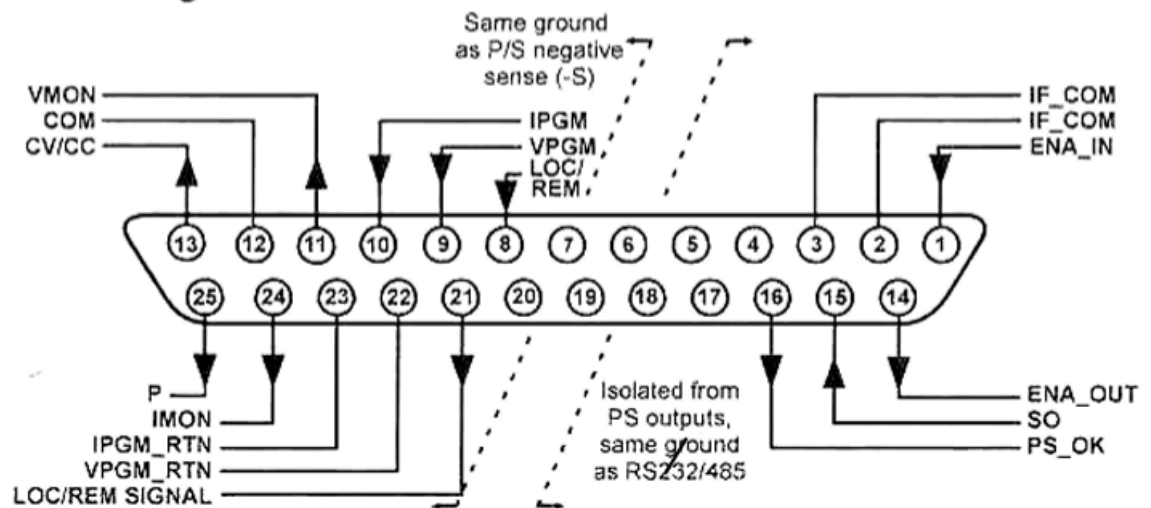
Set Up the Hardware :

- 1) Plug in the powersource into the DAQ (Plug it in at least 45 minutes before use for accurate readings).
- 2) Connect the USB into DAQ and laptop/computer
- 3) Insert thermocouple into a Channel not next to GRND
 - a) The Yellow end of the thermocouple wire connects into the positive/H channel port (for example CH0H)
 - b) The Red end of the thermocouple wire connects into the negative/L channel port (for example CH0L)
 - c) Make sure the opposite sides of the thermocouple aren't touching or entangled with each other.



4) Insert the Voltage and Current Probes

Fig.4-4: J1 connector terminals and functions

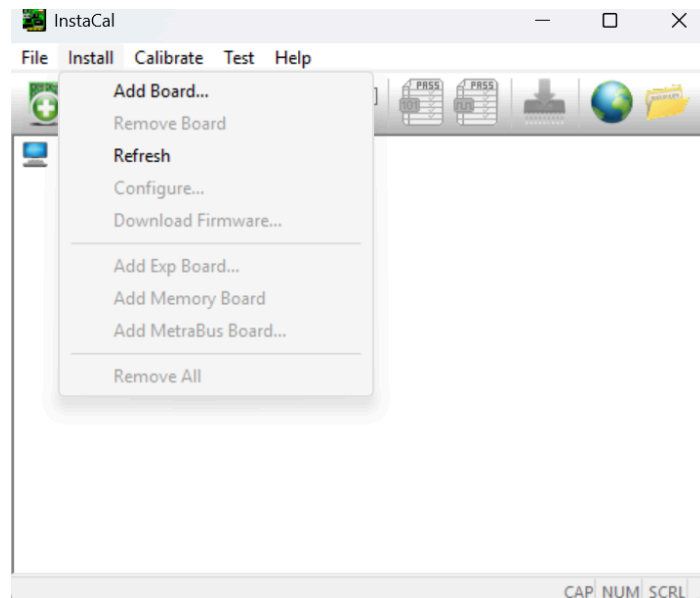


a) Insert one end of the voltage probe into port 11 then the other end into the positive/H channel port (for example CH2H)

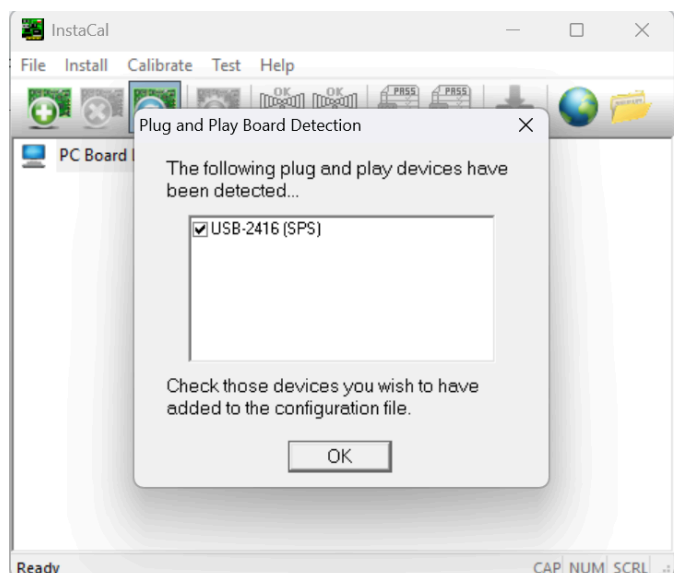
- b) Insert one end of the current probe into port 24 then the other end into the positive/H channel port (for example CH3H)
- c) Connect the ground probe into each of the negative/L channel ports (for example CH2L and CH3L) use a jumper cable if needed

Set Up InstaCal:

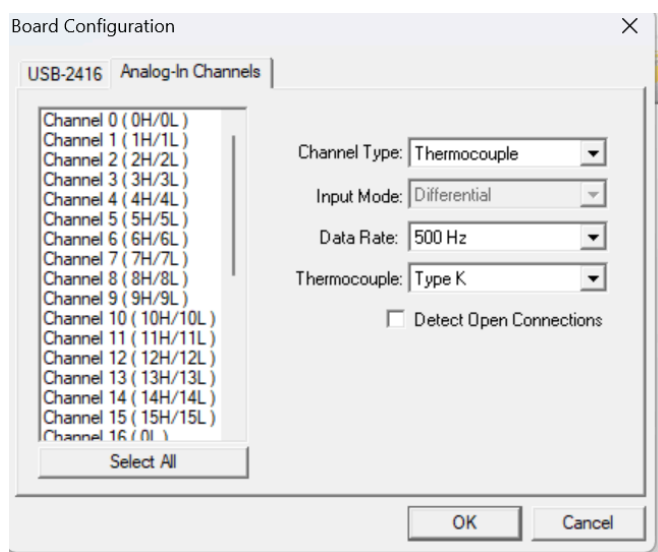
- 1) Open up InstaCal
- 2) Connect and set up the Board
 - a) If the board doesn't automatically connect then click on refresh



- b) You should get a pop-up that looks like this



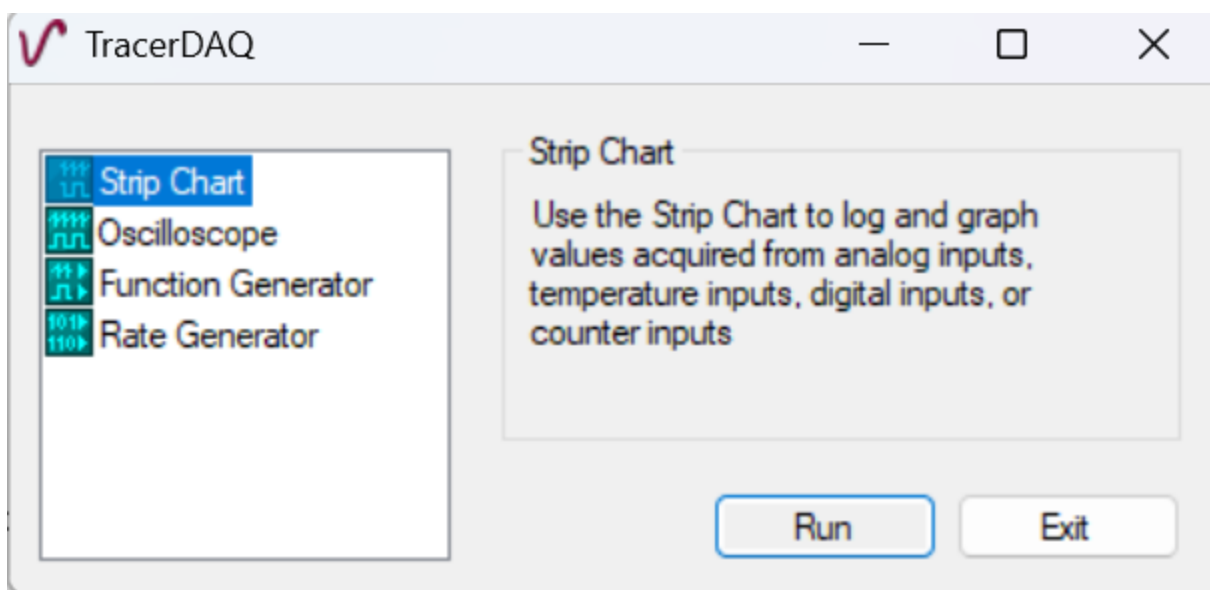
- c) Under **Install > Configure > Analog-in Channels** make sure your channel type is thermocouple and the correct thermocouple type is inputted.



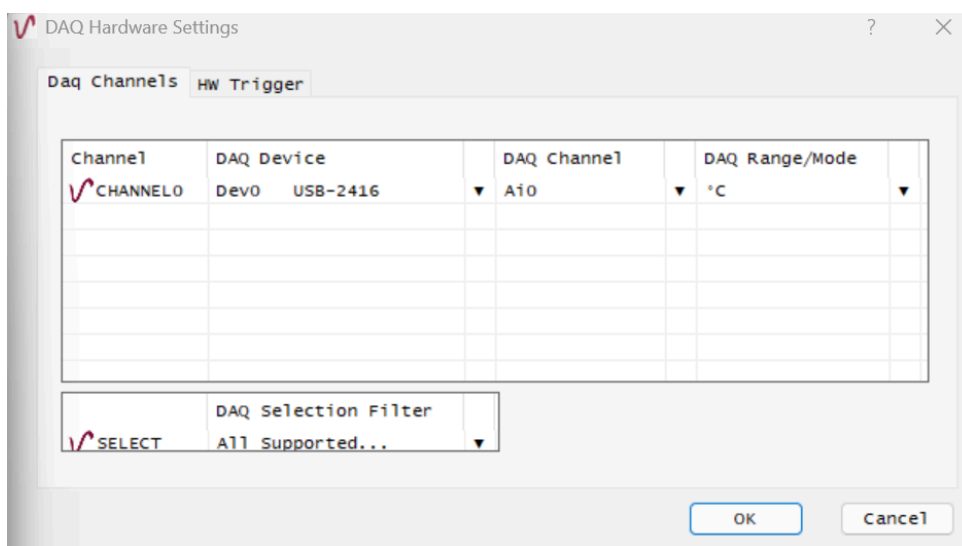
- d) Press **Test > Analog** and hold the end of the thermocouple in between two fingers (the reading should be around 33°C-35°C)

Set Up Tracer DAQ:

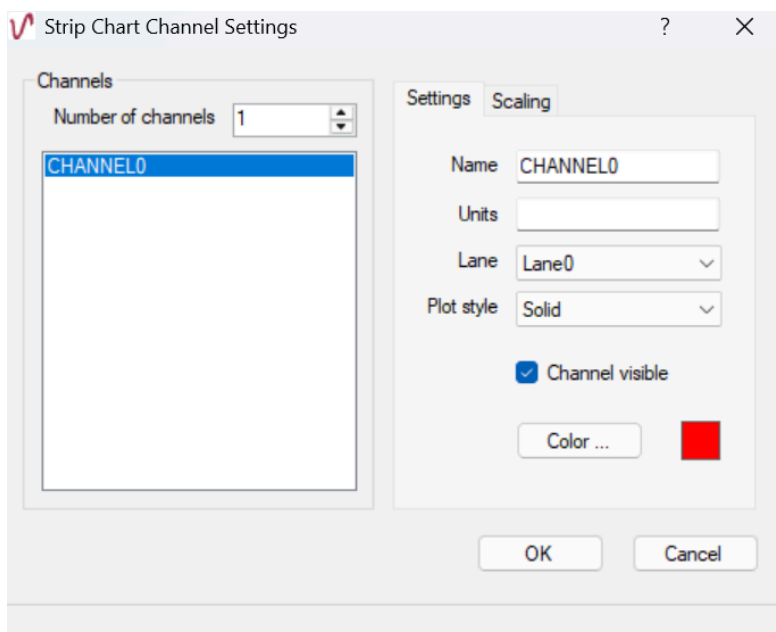
- 1) Select Strip Chart and Run



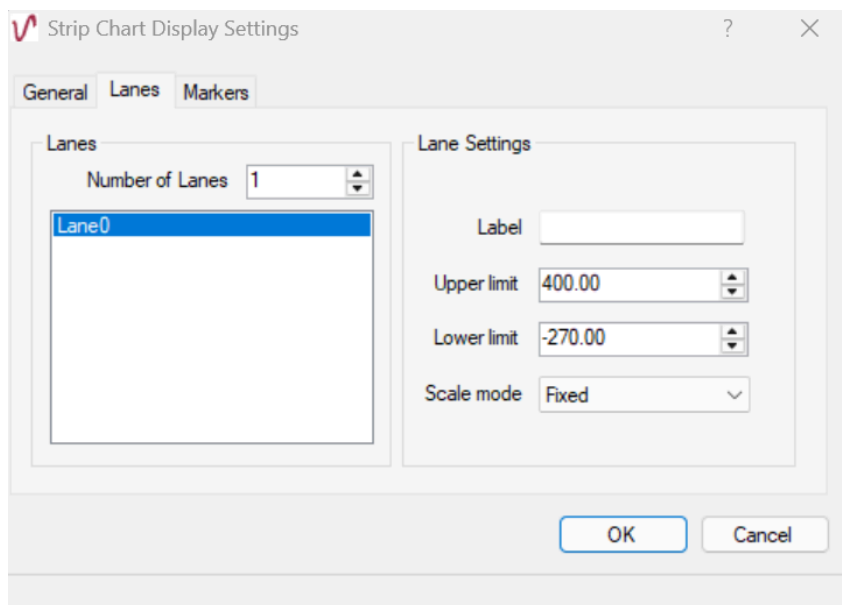
- 2) Press **Edit > > DAQ Hardware Settings** and select the correct DAQ Device and make sure the DAQ Range/Mode is in Celsius



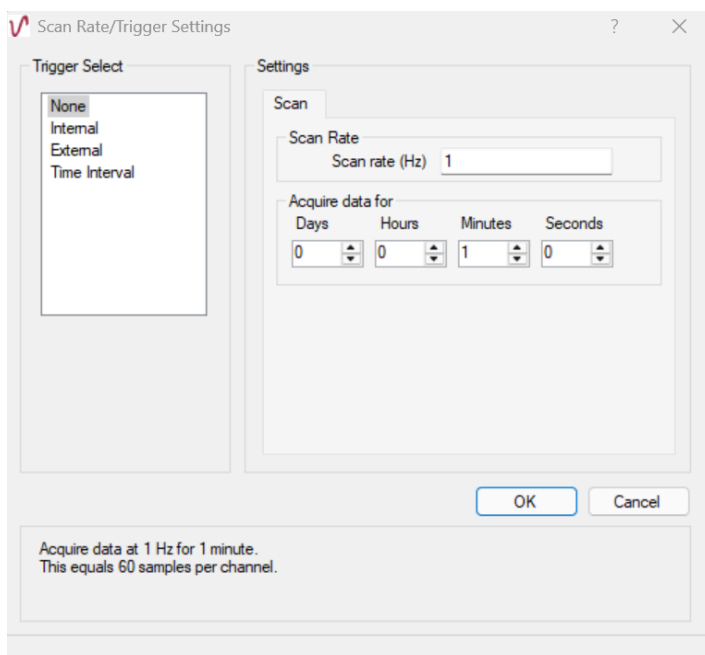
- 3) Press **Edit > Channel Settings** then adjust the Number of Channels to the correct amount you are using and make sure the Channels are referenced correctly and put into separate lanes



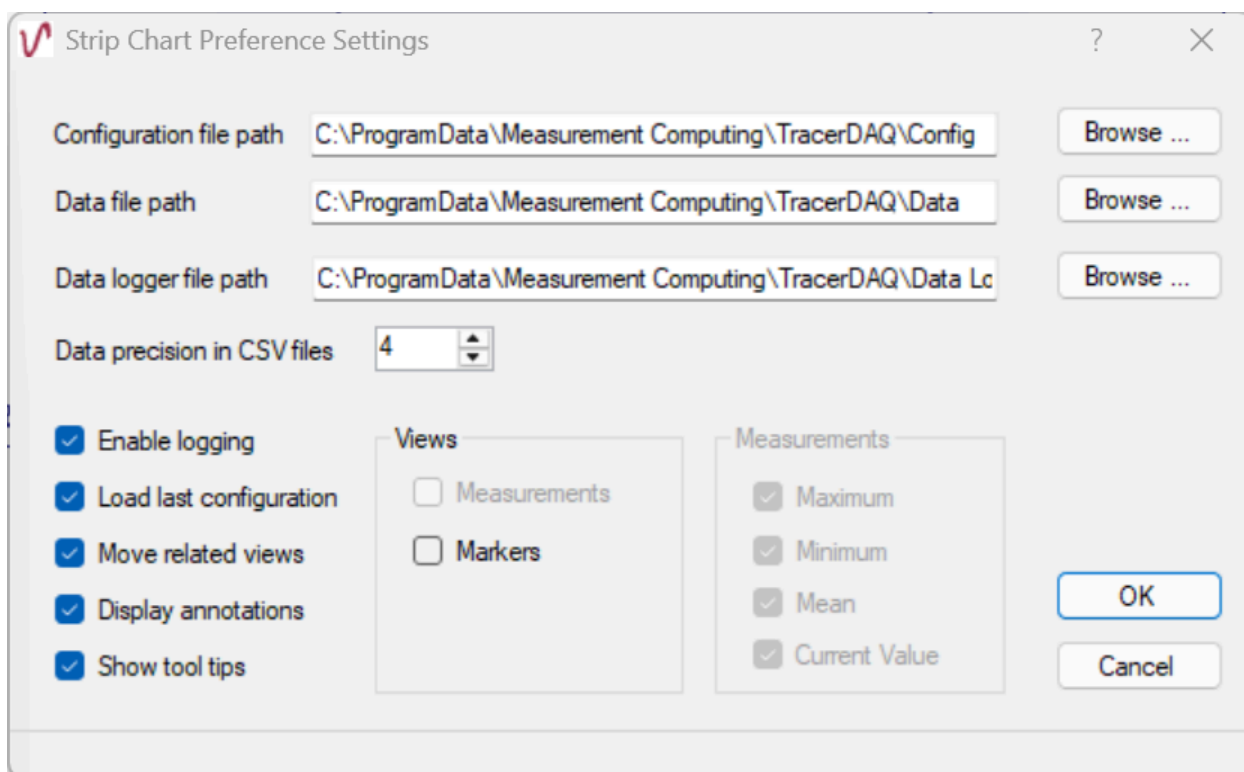
- 4) Press **Edit > Display Settings** then adjust the Number of Lanes to correlate to the correct amount of Channels you are using



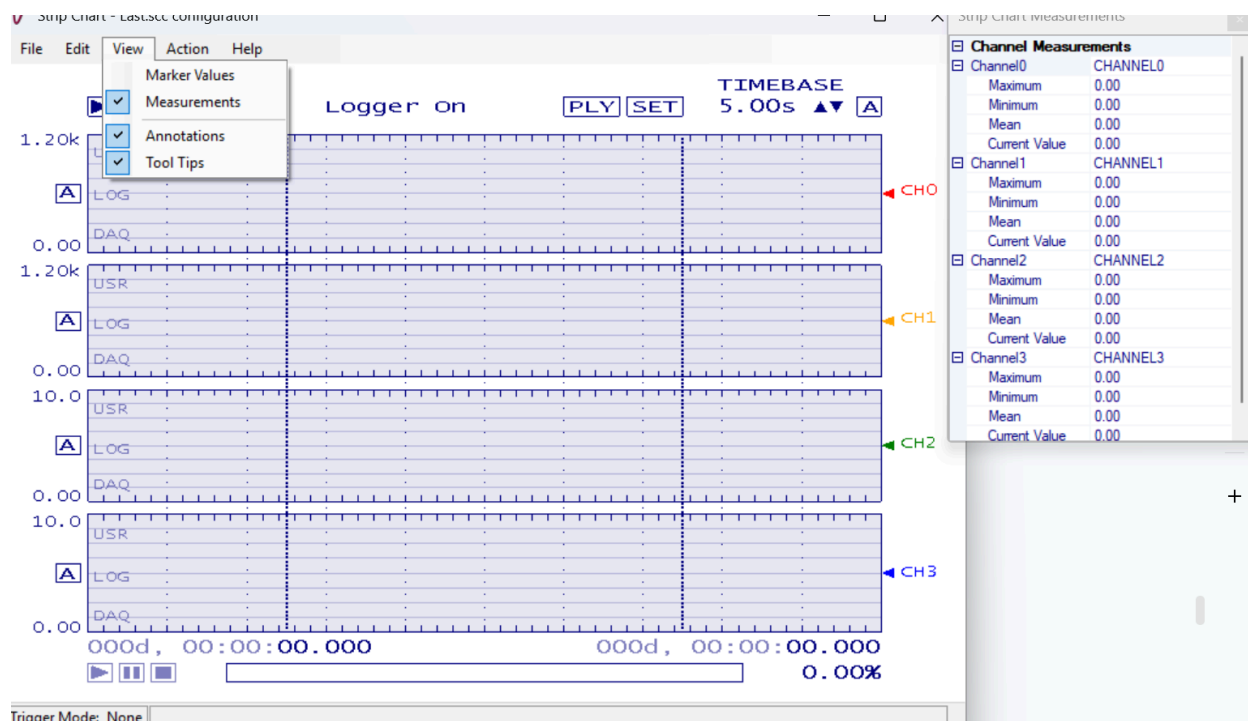
- 5) Press **Edit > Scan Rate/Trigger Settings** then choose how many data points you want recorded per second in the Scan Rate box



6) Press **File > Preferences** make sure Enable Logging is pressed



7) Press **View>Measurements** to have the real time measurements be displayed



- 8) Press **Action** and start your test run.
- 9) After the test run is completed press File > Save As name the file and choose the destination. **Then most importantly change the save as type from .sch to .csv.**

